Mathematics or the Verifier? No Detectable Gain from Verifier-Filtered Self-Training of Qwen2.5-1.5B, and a Per-Round Audit of One Verifier Defect

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Abstract. Self-training and reinforcement learning with verifiable rewards report large gains for small language models, but a rising score does not show that a model learned mathematics rather than what its verifier accepts; RL with random rewards has been shown to raise accuracy too. We study the cheapest form of the loop, generate–verify–select–train with offline LoRA SFT, on Qwen2.5-1.5B-Instruct with the rule-based verifier `math-verify`. On a held-out exam of 150 problems, at four learning rates on MATH-500 and two on GSM8K, no pass@1 or pass@k gain has a problem-level bootstrap interval excluding zero. The design could detect only gains of about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy (80% power); sample-accuracy intervals (a primary measure chosen after the runs) exclude gains above 2.2 points, a bound that omits training-seed variance. No arm, including training on reference solutions, improves on the untrained model, so this null concerns the recipe, not self-training in general. At the highest rate accuracy falls in every run, and in a post hoc measure matched on problem mix, finished solutions are 5.5–10.6 points less accurate. Our pre-set rule for verifier-style learning could fire only if accuracy rose. As a post hoc audit we track, per round, the share of the loop's own correct solutions that a storage defect in our pipeline rejects (about one in six). It falls by 1.5–2.6 points, which a falling `\boxed` share accounts for; on the problems the defect affects it stays near 95%. With 2–12 training examples per round that avoid the defect and no positive control, this finds no adaptation to the defect but cannot exclude it. A control that patches a rarer defect changes 0–4 training examples and cannot detect an effect; removing the filter also changes the training dose, and neither shows a consistent difference.

Keywords: self-training · self-improvement · verifier bias · mathematical reasoning · small language models · verifiable rewards · negative result.

1 Introduction

A common recipe for improving the mathematical reasoning of a small language model (SLM) *without human annotation* is to let the model generate its own solutions, keep those an automated verifier judges correct, and fine-tune on them, for several rounds. We call this the *Generate–Verify–Train* (GVT) loop; it underlies STaR [19], ReST [12], ReST-EM [11] and rejection-sampling fine-tuning [17]. But the verifier never reads the reasoning: a rule-based verifier such as `math-verify`[1] only extracts a final answer and compares it with the gold, and it rejects some correct answers written in forms it does not parse (Section 3.2). The training signal is "was the final string accepted", not "were the steps right".

The same gap runs through reinforcement learning with verifiable rewards (RLVR), which reports large gains at this scale: GRPO on a single training example lifts Qwen2.5-Math-1.5B from 36.0% to 73.6% on MATH-500 [14], a gain its authors partly trace to corrected output format, and similar recipes improve DeepSeek-R1-Distill-Qwen-1.5B [3] and open base models from 0.5B up [20]. Yet Shao et al. [9] find that GRPO with *random* rewards raises Qwen2.5-Math-7B by 21.4 points on MATH-500, against 29.1 with ground-truth rewards. A rising score therefore does not by itself show that a model has learned to solve more problems, rather than to surface behaviour it already had or to produce what its reward accepts. In RLVR, per-step false-negative rates on the policy's own rollouts have been tracked [4,16], and Egashira et al. report that a policy first unlearns the behaviour that triggers false negatives. This paper studies the question in the cheapest and most widely used form of the loop, offline rejection-sampling SFT, where a false reject only removes a training example, and applies the same kind of per-round false-reject measurement there, post hoc, for one defect of our pipeline.

We ask: *after several GVT rounds, is the model better at mathematics, or has it only learned to write what the verifier accepts?* We fixed four measures in advance: (i) pass@1; (ii) pass@k; (iii) output style, chiefly the share of generations containing `\boxed`; and (iv) per-problem transitions (problems *added* and *lost* between rounds). The pre-set reading rule was that "learning the verifier's style" requires pass@1 to rise, pass@k to stay flat and style to converge at once. In hindsight the rule is one-sided: it can only fire when accuracy rises (Section 5.1). Only these measures and the rule were fixed in advance. The higher learning rates, the controls, the round count and k at those rates, sample accuracy as primary measure, the termination measures and the re-scoring for a second verifier defect were all chosen after seeing earlier runs; we mark them as post hoc.

Our answer is negative on the first branch, within the power of the design: we detect no improvement on held-out problems at any learning rate we tried, but gains below about 10 points of pass@1 or 3.5–5.5 of sample accuracy could have been missed, and no arm of this recipe improves at all. On the second branch

[1] https://github.com/huggingface/math-verify, version 0.9.0 with antlr4 4.13.2 in this work.

the pre-set rule is silent. The post hoc audit of Section 5.3 finds no sign that the model learns to avoid one verifier defect, but it has little power to see such learning. What appears, most clearly at the highest rate, is lower accuracy on the solutions the model does finish and, clearly at only one of two seeds, more generations that run out of tokens. Two earlier readings did not hold up, and we report why: a pooled design whose sharp decline did not reproduce under isolation (Section 4.1), and an isolated design at a learning rate that changed the outputs little (Section 4.2).

Contributions.

- For a 1.5B model, LoRA SFT and one rule-based verifier, no gain is detectable on a held-out exam at four learning rates on MATH-500 or two on GSM8K, with problem-level bootstrap intervals for every comparison and the smallest effects the design could have detected. Because no arm improves, not even training on reference solutions, this is a null for the recipe.
- A per-round false-reject audit carried over from RLVR, where such rates are tracked per step [4,16], to offline SFT self-training (post hoc, Section 5.3), with the conditions it needs there. Overall, the share of correct solutions a gold-storage defect rejects falls by 1.5–2.6 points over rounds, which a falling `\boxed` share accounts for. Restricted to boxed answers or to the problems the defect affects, it is flat. The test is weak here: only 24–28 affected problems can be answered in an accepted form, the pool holds 2–12 examples per round that avoid the defect, and we have no positive control.
- An observation on style with its limits: the `\boxed` share falls under self-training in a way that cannot tell imitation of the pool from drift; the one strong style convergence occurs when training on reference solutions, with no verifier in the loop.
- A decomposition of an accuracy decline under a fixed token cap into a finished-accuracy part and a cap-share part, applied post hoc to what training at the highest rate changes: finished generations lose 5.5–10.6 points against the untrained model on the same problem mix (a descriptive measure that conditions on which problems finish), and the cap-hit share rises only at that rate, clearly at one of its two seeds.
- Design lessons: a verifier-patch control whose treatment changes 0–4 training examples cannot detect an effect, while a second, unvaried defect rejects about one in six correct solutions; and pooled and isolated designs differ in training dose as well as in the variables we meant to vary.

2 Related Work

Self-training and its limits. STaR [19] introduced the generate–filter–retrain loop, and ReST [12], ReST-EM [11] and RFT [17] formalise it; ReST-EM, which like us restarts from the base model each iteration, reports diminishing gains after the first round. None measures how the filter's errors depend on the form of the answer. Yue et al. [18] show that RL raises pass@1 while the base model wins

at large pass@k, and Teacher-Free Self-Training [13] concludes that self-training "amplifies but does not compound". Training repeatedly on model-generated data erodes distribution tails [10], and a narrow selector can accelerate collapse [8].

RL with verifiable rewards at small scale. RLVR with GRPO reports large gains near our model size: one training example lifts Qwen2.5-Math-1.5B on MATH-500 from 36.0% to 73.6% [14], about 7,000 examples lift DeepSeek-R1-Distill-Qwen-1.5B on AMC23 from 63% to 80% [3], and zero-RL works across ten open base models including Qwen2.5 from 0.5B [20]. These studies improve accuracy, which our SFT loop does not; Wang et al. separate out the part of their gain that is format correction, but the studies mostly do not ask what the rest consists of. Shao et al. [9] show that it need not be reward learning: random rewards raise Qwen2.5-Math-7B by 21.4 points on MATH-500, which they trace to amplified pre-existing behaviour, and the effect varies by model family. We do not compete with RLVR on accuracy; the audit of Section 5.3 asks of a loop whether the model has adapted to one verifier defect, and we apply it only to SFT.

Verifier error. TinyV [16] finds that rule-based verifiers reject many correct solutions; From Accuracy to Robustness [6] reports an average recall of 86%, and 0.95 for `math-verify` on their MATH subset. Imperfect Verifiers [1] models fixed false-accept and false-reject rates. Egashira et al. [4] find in RLVR that systematic false negatives behave like random noise whereas false positives can cause plateau or collapse. They also track the false-negative rate on the policy's own rollouts per training step and observe that the policy first unlearns the behaviour that triggers false negatives; TinyV likewise tracks the share of false negatives across GRPO steps. Our setting differs in paradigm (offline SFT with a fresh adapter per round) and task, and our controls cannot test their findings (Section 4.3).

Auditing self-improvement. Xu et al. [15] audit LoRA self-training of Qwen3-8B against a frozen control and identify seven measurement failures, including training-seed variance and a fixed token cap under style drift, that each invert a finding when uncontrolled. This is the study closest to ours; it is why we separate finished accuracy from termination (Section 4.5). Kim et al. [7] show that self-distillation on ground-truth solutions can cut accuracy while shortening responses; our train-on-gold control shows a similar surface signature. We found no multi-round study combining a measured format-biased rule-based verifier with a patched-verifier control, but since our patch proved too weak (Section 4.3) we do not claim this combination as the contribution. The per-round false-reject measurement of Section 5.3 is not new either: it is the per-step tracking of Egashira et al. and TinyV, applied here to offline SFT self-training and to a defect of our own pipeline. Appendix A gives the full comparison.

3 Setup

3.1 The GVT Loop

Each round t has four steps. *Generate:* the current model samples k solutions per problem. *Verify:* `math-verify` extracts the final answer and compares it with the gold. *Select:* accepted solutions are kept. *Train:* a *fresh* LoRA adapter is attached to the *base* model and trained on the kept solutions for one epoch. Round 0 is the untrained base model; the round-t model ($t \geq 1$) is the base plus an adapter trained on the pool accepted at round $t-1$. Adapters are not accumulated, so earlier rounds affect later ones only through the data they generate; conclusions about (non-)compounding below apply to this restart-from-base design, which gives compounding less room than continued training would.

3.2 The Verifier and Its Directional Bias

A format probe that re-renders gold answers in individually valid forms shows that `\dfrac`/`\tfrac` are rejected 100% of the time on both datasets (the command is not parsed). On real solutions (one public Qwen2.5-7B-Instruct solution per problem for 200 MATH-500 problems, judged correct by a scripted normalised comparison) the false-negative rate is 7.8% (10 of 128; Wilson interval 4.3–13.8%) on MATH-500 and 0% on GSM8K. Eight of the ten fail when the whole solution is verified but pass when only the boxed answer is; most are compound answers, so they are probably largely the gold-storage defect described below rather than extraction of a wrong token. The audit inspected only rejections, so it gives no false-positive rate. The verifier also cannot parse 51 of the 500 stored MATH-500 golds (10.2%); these problems always count as unsolved. That is a constant offset in counts within a partition, so it cancels in within-run and same-partition differences, but not in ratios or in levels compared across partitions (11–18 such problems per exam partition).

The patch. For the control we rewrote `\dfrac` and `\tfrac` to `\frac` before parsing, removing the 100%-rejection class and nothing else.

The gold-storage defect (post hoc). Our pipeline stores the gold without a delimiter, and `math-verify` applies its full LaTeX extraction only inside `\boxed{}` or `$...$`, so a correct boxed prediction can fail against a bare gold (e.g. the prediction `\boxed{\frac{4}{3}}` against the stored gold `\frac43`). Re-scoring the saved generations with a verifier V' that wraps the gold in `\boxed{}` when the prediction contains one shows that this defect rejects 15.5–18.4% of the generations V' accepts. It would enlarge the round-0 kept pool by about a fifth ($440 \to 542$), against 0–4 examples for the `\dfrac` patch, and it lowers absolute pass@1 by 4.7–8.0 points in self-training. It is bound to the problem, not to the answer: on the 54–62 problems where it ever fires, it rejects 94–98% of correct answers in every round, and 29–34 of these problems have a gold the original verifier cannot parse at all, so no answer form avoids it there. Re-scored, sample accuracy still falls in all five runs we re-scored; the original-verifier changes are 0.73–0.85 of the re-scored ones, an observed ratio that we do not use to rescale

other numbers or intervals. All tables keep the original verifier, the one the loop trained with. Section 5.3 uses this defect for the adaptation audit; Appendix B gives the full analysis.

3.3 Two Experimental Designs

Design P (pooled) loads both datasets (500 problems each) together and trains *one* adapter per round on the union of accepted solutions, with selection and evaluation on the same 1,000 problems. *Design I (isolated)* uses one dataset and one adapter per run, and holds out 30% of the 500 problems (150) as an `exam` set; generation runs on all 500 problems each round, but only the other 350 can enter the training pool, and all Design I pass rates are on the exam set. Table 9 summarises them.

Design I has three runs, all Qwen2.5-1.5B-Instruct, five rounds, $k = 8$, differing in one respect each: **Run A** (MATH-500, original verifier), **Run B** (MATH-500, patched verifier) and **Run C** (GSM8K, original verifier). Runs A and B were repeated at a second seed; since the seed also sets the train/exam partition, the repetition draws a different 150-problem exam set. Most of the later evidence comes from higher-rate runs introduced in Section 4.4; Table 1 lists every run. A "seed" sets the exam partition, the selection set and the training randomness at once; runs with the same seed share the partition. "Five rounds" means round 0 (the untrained model) and four adapters, each trained once on the previous round's pool; "three rounds" means round 0 and two adapters, so the round-2 model is a single adapter (about 195 optimizer steps for Run A at 5×10^{-4}) trained on the round-1 pool. All runs use one model family, Qwen2.5.

Table 1. All runs (Qwen2.5-1.5B-Instruct unless stated). Design P was run once per model. "Pool" is the round-0 kept pool (training examples; optimizer steps are half of it).

Design / run	Data	Rate	k, rounds	Seeds	Pool
P (0.5B, 1.5B)	GSM8K+MATH-500	10^{-5}	8, 5	one each	2,533, 3,962
I: A, B	MATH-500	10^{-5}	8, 5	0, 1	872, 833 (A)
I: C	GSM8K	10^{-5}	8, 5	0	1,901
I: A	MATH-500	10^{-4}	4, 3	1, 2, 3	424–455
I: A	MATH-500	2×10^{-4}	4, 3	1, 2	440, 424
I: A, B	MATH-500	5×10^{-4}	4, 3	1, 2	440, 424
I: no filter, gold	MATH-500	5×10^{-4}	4, 3	1, 2	1,400, 350
I: A replicate, 1024 limit	MATH-500	5×10^{-4}	4, 3	1	440
I: C	GSM8K	5×10^{-4}	4, 3	1	971

3.4 Models, Data, Training, Metrics

We use Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (Design P uses both, Design I the 1.5B model only) and the first 500 problems of the GSM8K test

split [2] and MATH-500, the 500-problem subset of MATH [5]. The prompt is a single user turn in the model's chat template: "Solve the math problem. Put the final answer in `\boxed{}`." followed by the problem. Generation samples at temperature 0.8 with a cap of 512 new tokens; the other decoding settings come from the checkpoint's generation configuration (top-p 0.8, top-k 20, repetition penalty 1.1). Training is LoRA SFT (`transformers` 5.0.0, `peft` 0.19.1, TRL 1.10.0 `SFTTrainer`; rank 16, $\alpha = 32$, on `q_proj` and `v_proj`, LoRA dropout and weight decay at the library defaults; batch size 2; one epoch; AdamW with linear learning-rate decay and no warmup; loss over the whole chat sequence, prompt included; fp16 on one Kaggle T4), with a training sequence limit of 512 tokens that includes the prompt. Because this limit equals the generation cap but also holds the prompt, any kept solution longer than about 512 tokens minus the prompt length is cut in training and loses its final answer and end-of-sequence token (24–26% of the round-0 pool; Section 4.5). All kept solutions are used, so the number of optimizer steps per round is half the pool size. The learning rate is 10^{-5} unless stated otherwise; this value was the default of our training code and was not tuned (Section 4.4). Metrics per round, on the exam set: pass@1 is the correctness of the *first* of the k samples (one draw per problem, so it carries sampling noise); pass@k; *sample accuracy*, the mean over all k samples; style counts over all generations (boxed, dfrac, dollar, ...); and per-problem added/lost counts between consecutive rounds, where a problem is *solved* in a round if at least one of its k samples is accepted. Added minus lost therefore equals $150 \times \Delta$pass@k, so these counts add only the gross churn behind the net change, not an independent test. pass@1 was fixed in advance; sample accuracy was not among the pre-set measures, but it estimates the same quantity from all k samples with less variance, so after the runs we chose it as the primary accuracy measure. Claims that rest on it, including the 2.2-point bound, are therefore post hoc in the choice of measure. "Correct" always means accepted by the verifier (preset `reward`), the same verifier that selects the training data.

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap (10,000 resamples of the 150 exam problems), which respects the clustering of the k samples within a problem. These intervals do *not* cover training-seed variance, which we see only across two or three seeds: the one same-configuration replicate pair we have ends 3.2 points of sample accuracy apart. We make no multiplicity correction, so we rest conclusions on patterns across runs and seeds, not on single intervals. Single proportions use Wilson intervals. *Detectable effects.* From the interval half-widths h in Table 5, a change would be detected with 80% power at the 5% level only if it exceeded about $2.8\,h/1.96$: about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy on MATH-500. A null in this paper means no gain of roughly that size. Details are in Appendix C.

Reproducibility. The analysis scripts named in the paper and the saved generations of every run are in our project repository. Token counts at generation

time, Design P's training logs and the per-sample outputs of the 1,536-token re-scorings were not saved. Every accuracy number in the paper is graded by the same biased verifier that selects the training data, except the re-scored figures of Appendix B.

4 Results

4.1 Design P: An Apparent Collapse

In the pooled design the 1.5B model declines severely and with acceleration (pass@1 50.0% → 18.6%, pass@8 72.6% → 48.6%; Table 2), on both datasets (Table 10); the 0.5B decline is small. The accepted pool contracts every round and `\boxed{}` falls from 5,608 to 2,693 of 8,000 generations. This design cannot be read causally: one adapter trains on both datasets' pools; it scores on the problems it selects from; and it trains on about 4.5× as many examples per round as the isolated MATH-500 run (3,962 against 872 at round 0). Selection-set scoring is unlikely to have masked a decline (in Design I the selection problems fall about as much as the exam at the higher rates, slightly less at 10^{-5}). Dose alone does not obviously fit the acceleration, since pools shrink while declines grow, but dose and data can interact. No run separates these explanations (Appendix D).

Table 2. Design P: pass@1 and pass@8 over five GVT rounds, pooled GSM8K + MATH-500, scored on the selection set. Round 0 is the untrained base model. Section 4.1 explains why these numbers cannot be attributed to any single cause.

Model	Metric	R0	R1	R2	R3	R4	Δ (R0→R4)
0.5B	pass@1	31.1%	31.9%	29.9%	28.9%	27.4%	−3.7 pt
0.5B	pass@8	60.5%	58.9%	56.1%	55.5%	54.9%	−5.6 pt
1.5B	pass@1	50.0%	46.2%	44.5%	32.4%	18.6%	**−31.4** pt
1.5B	pass@8	72.6%	71.5%	70.2%	63.6%	48.6%	**−24.0** pt

4.2 Design I at 10^{-5}: Small Declines, a Larger One on GSM8K

On the held-out exam (Fig. 1, Table 3) pass@8 ends within 4 points of round 0 in every run. On MATH-500, pass@1 is 2.7–4.0 points lower at round 4, with every interval including zero (Table 5); sample accuracy falls 1.2–3.2 points, with an interval excluding zero only for B at seed 0 (A at seed 0 reaches 0.0), and matched finished accuracy 1.1–4.2, with intervals that only just exclude zero for A and B at seed 0, which are close to one replicate. We read this as a possible small decline, not an established one; what we do not see is a gain of the size the design could detect. Run C (GSM8K) declines more: pass@1 falls 12.7 points ($[-21.3, -4.0]$), almost all in finished accuracy, but it took about twice Run A's optimizer steps, so we do not attribute this to the dataset alone. At this rate

the training loss falls, but on MATH-500 two coarse output statistics, length and cap-hit share, barely change; we did not measure a finer one, so this is weak evidence that the adapter did little, and a flat curve there does not show that GVT is stable (Appendix E).

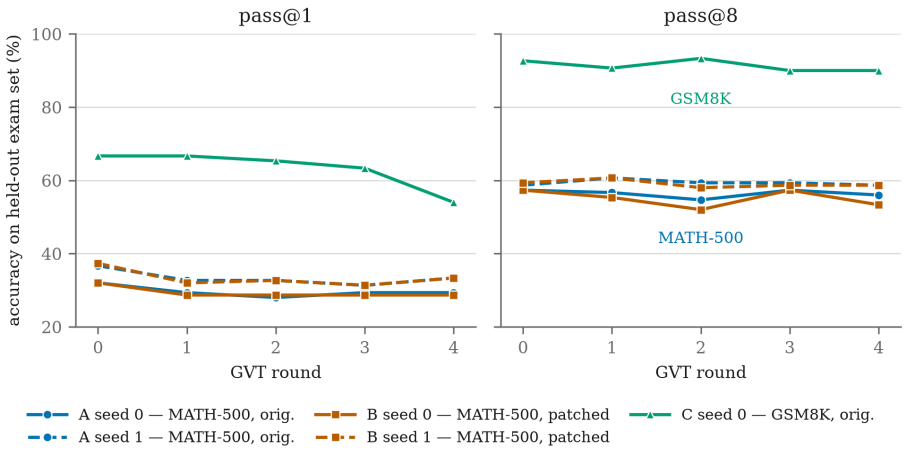


Fig. 1. Design I at 10^{-5}: pass@1 (left) and pass@8 (right) on the held-out exam set, by round. Colour identifies the run, line style the seed, marker the run again, so the figure survives greyscale printing. Runs A and B track each other closely at both seeds. pass@8 stays within a few points in every run. The one large visible movement is Run C's pass@1 in the final round.

Table 3. Design I: pass@1 and pass@8 on the held-out exam set (150 problems), Qwen2.5-1.5B, learning rate 10^{-5}. Runs A and B differ only in the `\dfrac`/`\tfrac` patch. Each seed draws its own exam partition, so round-0 values differ between seeds.

Run	Metric	R0	R1	R2	R3	R4	Δ
A, seed 0 (MATH-500, orig.)	pass@1	32.0%	29.3%	28.0%	29.3%	29.3%	−2.7 pt
A, seed 0 (MATH-500, orig.)	pass@8	57.3%	56.7%	54.7%	57.3%	56.0%	−1.3 pt
A, seed 1 (MATH-500, orig.)	pass@1	36.7%	32.7%	32.7%	31.3%	33.3%	−3.3 pt
A, seed 1 (MATH-500, orig.)	pass@8	58.7%	60.7%	59.3%	59.3%	58.7%	0.0 pt
B, seed 0 (MATH-500, patched)	pass@1	32.0%	28.7%	28.7%	28.7%	28.7%	−3.3 pt
B, seed 0 (MATH-500, patched)	pass@8	57.3%	55.3%	52.0%	57.3%	53.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@1	37.3%	32.0%	32.7%	31.3%	33.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@8	59.3%	60.7%	58.0%	58.7%	58.7%	−0.7 pt
C, seed 0 (GSM8K, orig.)	pass@1	66.7%	66.7%	65.3%	63.3%	54.0%	−12.7 pt
C, seed 0 (GSM8K, orig.)	pass@8	92.7%	90.7%	93.3%	90.0%	90.0%	−2.7 pt

4.3 The Patch Control Could Not Detect an Effect

Runs A and B differ only in the `\dfrac`/`\tfrac` patch. At round 4, B minus A is -0.7 [-3.3, $+2.0$] points of pass@1 at seed 0 and $+0.0$ [-2.7, $+2.7$] at seed 1. This null is uninformative: `\dfrac` appears in 0.50–0.73% of generations, so the kept pools differ by 3–4 examples (under 0.5%), and at 10^{-5} the adapter changes the outputs little anyway. Removing one rare format defect had no detectable effect, which a prevalence count alone predicts; the frequent gold-storage defect is present in both arms. The pre-filter share of problems on which all k samples agree falls in each of the four runs where we measured it (Design P 1.5B and the seed-0 10^{-5} runs, over all 500 items), but it is mechanically tied to accuracy, so we draw no conclusion from it (Appendix F).

4.4 Higher Learning Rates

Because 10^{-5} changed the MATH-500 outputs little, we reran Run A at 10^{-4} (three seeds), 2×10^{-4} (two) and 5×10^{-4} (two, with Run B), with three rounds and $k = 4$ (Table 4). These runs therefore differ from the 10^{-5} runs in k, round count, pool size (and so optimizer steps) and the pass@k metric as well as in rate, so only the comparison among 10^{-4}, 2×10^{-4} and 5×10^{-4} isolates the rate. *At* 5×10^{-4} *every run declines:* pass@1 falls 4.0–11.3 points and the sample-accuracy intervals exclude zero in all four A/B runs, though same-seed A and B are closer to two replicates than four. *The decline may grow with the rate on one seed:* at seed 1 sample accuracy falls 1.5, 3.5 and 8.3 points across the three rates, at seed 2 an almost flat 3.7, 3.8 and 4.7. The cap-hit share rises only at 5×10^{-4}. *The patch has no consistent effect:* B minus A in round-2 pass@1 is $+5.3$ at seed 1 and -7.3 [-14.0, -0.7] at seed 2, a sign flip from pools that differ by 0–3 examples, which illustrates trajectory noise. *Style does not separate imitation from drift:* finished exam generations start more boxed than the verifier-accepted pool and fall below it as the pool falls too, which imitation of the pool and drift would both produce.

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem (1,400 examples per round), removing the verifier; the *train-on-gold* run trains on the 350 reference solutions every round (Table 6). Neither is a single-variable control: no-filter takes 3–4× Run A's optimizer steps, train-on-gold about the same. Removing the filter is never better than keeping it beyond noise and is worse on pass@4 at both seeds (round-2 differences -2.0 and -5.3 points, intervals including zero), but the dose difference means this cannot be attributed to the filter; the no-filter pool is also truncated far more often in training (62.4% against 24.5% at round 0). Training on reference solutions is followed by a much larger decline (pass@4 down about 21 points at both seeds) while the model adopts the reference style (`$`-delimited spans 5% → 98%). Post hoc checks bound verifier error from below only (about 2 points of sample accuracy among boxed answers), and the 14% of round-2 generations without `\boxed` are not covered, so we think this fall is unlikely to be mainly verifier

Table 4. Run A (MATH-500, original verifier, $k = 4$, three rounds) across learning rates and seeds: pass@1, pass@4 and sample accuracy (600 exam generations) at round 0 $\rightarrow$ round 2, the share of round-2 generations that hit the 512-token cap (decoded text re-tokenised to ≥ 508 tokens; round 0 is 43.0–43.2%, 48.2% at seed 3), and problems added/lost summed over rounds 1–2. The two 5×10^{-4} rows are the A rows of Table 11.

Rate	Seed	pass@1	pass@4	Sample acc.	Cap-hit (R2)	Added/lost
10^{-4}	1	33.3 $\rightarrow$ 34.0	50.0 $\rightarrow$ 49.3	33.7 $\rightarrow$ 32.2	42.7%	20/21
10^{-4}	2	31.3 $\rightarrow$ 28.0	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.2	42.8%	25/34
10^{-4}	3	29.3 $\rightarrow$ 26.7	40.0 $\rightarrow$ 41.3	28.5 $\rightarrow$ 27.5	45.7%	16/14
2×10^{-4}	1	33.3 $\rightarrow$ 31.3	50.0 $\rightarrow$ 46.7	33.7 $\rightarrow$ 30.2	40.8%	23/28
2×10^{-4}	2	31.3 $\rightarrow$ 25.3	51.3 $\rightarrow$ 45.3	30.8 $\rightarrow$ 27.0	39.7%	21/30
5×10^{-4}	1	33.3 $\rightarrow$ 23.3	50.0 $\rightarrow$ 42.7	33.7 $\rightarrow$ 25.3	46.2%	22/33
5×10^{-4}	2	31.3 $\rightarrow$ 27.3	51.3 $\rightarrow$ 44.7	30.8 $\rightarrow$ 26.2	51.2%	26/36

Table 5. Change from round 0 to the last round (percentage points) with 95% paired problem-level bootstrap intervals over the 150 exam problems (10,000 resamples; `experiments/viec3/bootstrap_ci.py`). Last round is R4 at 10^{-5} ($k = 8$) and R2 elsewhere ($k = 4$). "Finished, matched" is the last-round accuracy of generations below the 512-token cap minus the untrained model's accuracy on the same problem mix (Section 4.5; 2,000 resamples, `experiments/viec3/review_cpu/capanat.py`). The intervals do not include training-seed variance. Recomputed values match the reported ones except six round values that differ by one problem (Section 3.4), e.g. B, seed 1 at 10^{-5}, whose round-4 pass@8 is 58.0% against 58.7%; the matched column uses the separate re-scoring described there.

Run, seed	Δpass@1	Δpass@k	Δsample acc.	Finished, matched
Learning rate 10^{-5}				
A, 0	-2.7 [$-9.3, 4.0$]	-1.3 [$-8.0, 5.3$]	-2.4 [$-4.8, 0.0$]	-3.8 [$-7.2, -0.5$]
A, 1	-3.3 [$-10.0, 3.3$]	$+0.0$ [$-6.0, 6.0$]	-1.5 [$-4.1, 1.0$]	-2.3 [$-5.9, 1.2$]
B, 0	-3.3 [$-10.0, 3.3$]	-4.0 [$-10.7, 2.7$]	-3.2 [$-5.9, -0.6$]	-4.2 [$-7.8, -0.9$]
B, 1	-4.0 [$-10.7, 2.7$]	-1.3 [$-7.3, 4.7$]	-1.2 [$-3.7, 1.3$]	-1.1 [$-4.6, 2.6$]
C, 0	-12.7 [$-21.3, -4.0$]	-2.7 [$-7.3, 2.0$]	-13.1 [$-16.7, -9.8$]	-12.8 [$-16.5, -9.4$]
Learning rate 10^{-4}				
A, 1	$+0.7$ [$-6.7, 8.0$]	-0.7 [$-6.0, 4.7$]	-1.5 [$-5.3, 2.2$]	-4.3 [$-9.6, 1.1$]
A, 2	-3.3 [$-9.3, 2.7$]	-6.0 [$-12.7, 0.7$]	-3.7 [$-7.3, -0.2$]	-3.5 [$-9.2, 2.3$]
A, 3	-2.7 [$-10.0, 4.0$]	$+1.3$ [$-4.7, 7.3$]	-1.0 [$-4.3, 2.2$]	-0.6 [$-6.0, 4.7$]
Learning rate 2×10^{-4}				
A, 1	-2.0 [$-8.0, 4.0$]	-3.3 [$-10.0, 2.7$]	-3.5 [$-7.2, 0.0$]	-4.0 [$-8.9, 0.5$]
A, 2	-6.0 [$-12.7, 0.7$]	-6.0 [$-12.7, 0.7$]	-3.8 [$-7.5, -0.2$]	-4.7 [$-10.1, 1.0$]
Learning rate 5×10^{-4}				
A, 1	-10.0 [$-16.7, -3.3$]	-7.3 [$-14.0, 0.0$]	-8.3 [$-12.3, -4.5$]	-9.1 [$-14.2, -4.2$]
A, 2	-4.0 [$-11.3, 3.3$]	-6.7 [$-13.3, 0.0$]	-4.7 [$-8.2, -1.2$]	-5.5 [$-11.7, 0.1$]
B, 1	-4.7 [$-12.0, 2.7$]	-2.0 [$-8.7, 4.7$]	-6.7 [$-10.5, -2.8$]	-6.8 [$-12.4, -1.0$]
B, 2	-11.3 [$-17.3, -5.3$]	-11.3 [$-18.0, -4.7$]	-9.2 [$-12.8, -5.5$]	-5.9 [$-10.9, -1.0$]
No filter, 1	-6.0 [$-12.7, 1.3$]	-9.3 [$-16.7, -2.7$]	-8.2 [$-12.2, -4.3$]	-10.6 [$-16.4, -5.5$]
No filter, 2	-8.7 [$-16.0, -1.3$]	-12.0 [$-19.3, -4.7$]	-10.2 [$-14.2, -6.3$]	-6.6 [$-11.4, -2.0$]
Gold, 1	-18.0 [$-26.7, -10.0$]	-20.7 [$-28.7, -12.7$]	-19.7 [$-25.3, -14.2$]	-25.6 [$-32.1, -18.9$]
Gold, 2	-21.3 [$-28.7, -14.0$]	-20.7 [$-29.3, -12.7$]	-18.7 [$-23.7, -13.7$]	-25.9 [$-31.7, -20.0$]
C, 1	-22.0 [$-29.3, -15.3$]	-9.3 [$-15.3, -3.3$]	-15.3 [$-19.8, -11.2$]	-12.1 [$-16.6, -7.9$]

error but cannot bound it from above. That no arm, this one included, improves on the untrained model means we have no positive reference: the nulls above may reflect the recipe (LoRA on q/v projections, loss over the prompt, a 512-token limit) rather than self-training. Run C at 5×10^{-4} (GSM8K, one run) falls 22.0 points of pass@1, from both less accurate and more unfinished solutions (Appendix G).

Table 6. Controls at 5×10^{-4} (MATH-500, $k = 4$, three rounds), on the exam partition of Run A at the same seed. Pool is the number of training examples per round; optimizer steps are half of it.

Run	Pool	pass@1			pass@4		
		R0	R1	R2	R0	R1	R2
Seed 1							
A (verifier-filtered)	440→326	33.3	32.7	23.3	50.0	49.3	42.7
No filter	1400	33.3	26.7	27.3	50.0	49.3	40.7
Train on gold	350	33.3	18.0	15.3	50.0	33.3	29.3
Seed 2							
A (verifier-filtered)	424→350	31.3	31.3	27.3	51.3	46.0	44.7
No filter	1400	31.3	34.7	22.7	51.3	48.7	39.3
Train on gold	350	31.3	8.7	10.0	51.3	28.7	30.7

4.5 Anatomy of the Decline

This subsection is post hoc and descriptive. Sample accuracy mixes whether a finished solution is right with whether the model finishes within the 512-token cap at all. Re-tokenising the saved text, we call a generation *cap-hit* at ≥ 508 tokens. The untrained model already hits the cap in 41–48% of MATH-500 generations, so the cap binds from the start. An earlier character-length proxy for unfinished generations overstated termination, because drifting prose crosses its threshold within the same token budget. At 5×10^{-4}, compared with the untrained model on the same problem mix, finished generations lose 5.5–10.6 points, with intervals excluding zero in five of six runs (the six include two same-seed A/B pairs, which are close to replicates), while the cap-hit share rises 2–14 points. The matched measure assumes that a problem's round-0 finished accuracy is the right baseline for the samples that still finish, and it conditions on which problems finish after training, so it is descriptive. A three-part decomposition attributes 12–30% of the decline to the cap share in the three original seed-1 runs (and −2 and 5% in the replicate and the 1,024-limit run) and 52–72% at seed 2 (Table 7). At 10^{-4} and 2×10^{-4} the cap-hit share does not rise and finished accuracy carries the whole change. Train-on-gold is different: responses shorten by more than half and finished accuracy collapses. Retraining without truncated examples changes little within the noise of single runs; re-scoring adapters at a 1,536-token cap

leaves their deficit in place, and the character proxy there points to generations that do not stop rather than to less accurate finished ones (Appendix H).

Table 7. Decomposition of the change in sample accuracy (points, round 0 → last round, MATH-500 exam unless stated) into finished-accuracy, cap-share and cap-accuracy parts, with the change in cap-hit share. Intervals: paired problem-level bootstrap, 2,000 resamples. The share is the finished part over the whole change.

Run, seed	Δcap-hit	Δacc.	Finished	Cap-share	Cap-acc.	Fin. share
5×10^{-4}						
A, 1	+3.0 [−1.7, 7.7]	−8.3	−6.0	−1.3	−1.0	73% [43, 108]
B, 1	+4.7 [0.2, 9.5]	−6.7	−4.2	−2.0	−0.5	62% [21, 100]
No filter, 1	+2.3 [−2.5, 6.8]	−8.2	−6.0	−1.0	−1.1	74% [46, 106]
A, 2	+8.2 [3.8, 12.3]	−4.7	−2.7	−3.4	+1.5	59% [−11, 114]
B, 2	+11.0 [6.2, 16.0]	−9.2	−3.6	−4.8	−0.8	39% [6, 64]
No filter, 2	+14.2 [9.3, 18.8]	−10.2	−3.5	−6.1	−0.6	34% [5, 57]
A replicate, 1	−0.2 [−4.5, 4.3]	−5.2	−4.6	+0.1	−0.7	89% [41, 162]
A, 1024 limit, 1	+0.7 [−3.7, 5.2]	−5.7	−4.8	−0.3	−0.5	85% [42, 169]
C (GSM8K), 1	+11.2 [7.8, 14.7]	−15.3	−7.8	−6.6	−0.9	51% [30, 66]
2×10^{-4} *and* 10^{-4}						
A, 2×10^{-4}, 1	−2.3 [−6.7, 2.3]	−3.5	−4.3	+1.0	−0.2	124%
A, 2×10^{-4}, 2	−3.3 [−7.7, 1.0]	−3.7	−5.2	+1.4	+0.1	142%
A, 10^{-4}, 1	−0.5 [−4.5, 3.7]	−1.8	−3.1	+0.2	+1.0	169%
A, 10^{-4}, 2	−0.2 [−4.3, 4.0]	−3.7	−3.4	+0.1	−0.3	93%
A, 10^{-4}, 3	−2.5 [−6.3, 1.3]	−1.0	−1.4	+1.1	−0.7	139%

5 Discussion

5.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. No run raises pass@1 or pass@k by an amount whose interval excludes zero, at any rate, and the sample-accuracy intervals exclude gains above about 2.2 points. The bound is loose: it is read off observed intervals rather than a pre-specified equivalence test, uses a measure chosen after the runs (on pass@1 and pass@k the upper ends reach +8.0 and +7.3), and omits training-seed variance. On the pre-set measures the design could detect only gains of about 10 points of pass@1 and 8–10 of pass@k, so "no gain" means no gain of that size. In every MATH-500 run but one the problems lost equal or outnumber those gained, which restates the sign of the pass@k change rather than adding evidence. Because no arm improves, including training on reference solutions, the null may be a property of this LoRA recipe rather than of self-training.

Does it learn to write what the verifier accepts? The data cannot settle this. The pre-set pattern requires pass@1 to rise, which never happens detectably (the largest rise is +0.7 [−6.7, +8.0]), so the rule's silence is not evidence against verifier-style learning. The `\boxed` share falls in a way imitation and drift would both produce. The post hoc audit of Section 5.3 finds no sign of adaptation to the gold-storage defect, but it is too weak to rule adaptation out.

What happened to the pooled collapse? It does not reproduce when each dataset has its own adapter and is scored on held-out problems. Because the pooled design also trained on about 4.5× more examples per round, we cannot say whether pooling, selection-set scoring or dose produced it.

5.2 Interpretation

An earlier version explained the Design P curves by a self-reinforcing shrinking pool shaped by the biased verifier. The data neither support nor rule it out: the patch tests how strong a rare bias is, not whether a frequent bias compounds; the no-filter control declines without any verifier but with more dose. The defensible statement is narrow: *removing a format defect present in under* 1% *of generations, or removing the filter entirely, made no consistent difference at the resolution of this setup.* The frequent gold-storage defect was never varied, so whether systematic false negatives are harmless in general, as Egashira et al. find for RLVR, our data cannot say; and the no-filter control admits unstructured wrong answers, not the systematic false positives they find harmful, so it does not test that side either. Practices our experience suggests, none of them tested here, are listed in Appendix I.

5.3 A Proposed Audit for GVT with a Biased Verifier

Our pre-set reading rule failed in a predictable way: it tied "learning the verifier's style" to rising accuracy, so it could not speak once accuracy fell. From the analyses we added after the runs we propose four checks for self-training with a rule-based verifier. All four were defined post hoc here; we offer them as a protocol for future studies, not as tested methods.

(i) A per-round false-reject audit. Fix a repaired verifier V' that corrects a known defect class of the training verifier V. At each round t, on the loop's own generations, compute

$$r_t = \frac{\#\{\text{accepted by } V' \text{ but rejected by } V\}}{\#\{\text{accepted by } V'\}}.$$

If the model learns to write what V accepts, r_t should fall over rounds, whatever happens to accuracy. This is the per-step false-negative tracking of RLVR studies [4,16] applied to offline SFT. Our data show that overall r_t is not enough on its own (Table 8; `experiments/viec3/review_cpu/rt_audit.py`). For the gold-storage defect it falls in all three self-training runs we re-scored, by 2.6, 2.6 and 1.5 points, with intervals reaching or just touching zero. But V' differs from V only on boxed predictions, so unboxed correct answers enter only the denominator, and the fall matches the falling `\boxed` share among correct answers (91 → 76% and 93 → 78% at 5×10^{-4}). On boxed answers alone r_t is flat (changes −0.4 to +0.7 points). The defect is also bound to the problem: it fires on only 54–62 of the 500 problems, where it rejects 94–98% of correct answers in every round, and 29–34 of them have a gold that V cannot parse,

so no answer form could escape it there. On the 24–28 affected problems where some form could, r_t moves by -5.5, -1.6 and -0.2 points, with intervals 10–26 points wide. A model adapting to the defect would have to write these answers unboxed, and the training pool holds only 2–12 examples per round from these problems that do so. Under SFT the pressure to adapt is therefore weak, and we have no positive control showing the audit would detect adaptation that did occur. We conclude only that we see no adaptation to this defect, in one defect class, three runs and three to five rounds. Train-on-gold, with no verifier in the loop, gives r_t rises of 2.1–3.3 points as its outputs become more boxed, a reminder that r_t moves with format for reasons unrelated to the verifier. For the test to be informative, r_t should be reported on the subset of answers and problems where V and V' can differ and an accepted form exists, with a count of the training examples that could teach the avoidance.

Table 8. The false-reject audit for the gold-storage defect (post hoc), in percent (all 500 problems), round 0 → last round, with the change and a 95% problem-level bootstrap interval (2,000 resamples). "Boxed only" restricts to predictions containing `\boxed`; "affected, parseable" to problems where the defect ever fires and V can parse the gold. Pool: examples per round from affected problems that V accepts (and so avoid the defect).

Run	r_t subset	R0	Last	Δ	95% interval	Pool
A, 5×10^{-4}, seed 1	all	18.3	15.7	−2.6	$[-5.3, 0.3]$	3–5
	boxed only	20.0	20.7	+0.7	$[-2.9, 4.3]$	
	affected, parseable	92.8	87.2	−5.5	$[-20.1, 6.3]$	
A, 5×10^{-4}, seed 2	all	18.1	15.5	−2.6	$[-5.2, 0.1]$	2–3
	boxed only	19.4	19.8	+0.4	$[-2.8, 3.4]$	
	affected, parseable	95.1	93.5	−1.6	$[-14.7, 9.5]$	
A, 10^{-5}, seed 0	all	18.3	16.8	−1.5	$[-2.9, 0.0]$	9–12
	boxed only	19.9	19.5	−0.4	$[-2.0, 1.2]$	
	affected, parseable	92.3	92.0	−0.2	$[-4.9, 4.6]$	
Gold, seed 1	all	18.2	21.5	+3.3	$[-1.2, 8.1]$	—
	boxed only	20.0	22.8	+2.8	$[-2.0, 7.7]$	
	affected, parseable	95.5	94.7	−0.7	$[-9.1, 6.2]$	
Gold, seed 2	all	18.0	20.2	+2.1	$[-2.2, 6.7]$	—
	boxed only	19.4	21.1	+1.7	$[-3.3, 6.5]$	
	affected, parseable	95.1	97.6	+2.5	$[-4.8, 11.9]$	

(ii) Decompose declines under a fixed cap. Report the change in sample accuracy split into a finished-accuracy part and a cap-share part (Table 7). Here the split is descriptive and its balance differs by partition: at 5×10^{-4} the cap share accounts for 12–30% of the decline in the three original seed-1 runs and 52–72% at seed 2.

(iii) Check the dose of a control before running it. Count how many training examples the treatment changes. Our `\dfrac` patch changed 0–4 examples, while the unvaried gold-storage defect would have changed about a fifth of the pool (440 → 542 at round 0); the first number alone shows the patch control could not detect an effect.

(iv) Match training dose across designs. Our pooled design trained on about 4.5× more examples per round than the isolated one, so its sharp decline cannot be attributed to pooling.

Check (i) in RLVR. There a false reject gives the correct answer zero reward, a direct pressure that SFT lacks, and Egashira et al. already observe the policy unlearning behaviour that triggers false negatives. The audit needs only checkpoints, their generations on fixed problems and a repaired verifier, so it can be run on public RLVR checkpoints, with the conditioning above; we have not done so.

6 Limitations

Unequal training doses confound every cross-arm comparison. *The controls are underpowered:* the patch changes 0–4 training examples, and the no-filter and train-on-gold controls have two seeds each. *Intervals cover only problem draw and sampling,* not training-seed variance; several conditions have one seed, and no multiplicity correction is applied. *The learning-rate grid is coarse,* with three rounds and $k = 4$, and it changes k, rounds, dose and metric along with the rate. *The power is limited:* with 150 exam problems only gains of about 10 points of pass@1 or 3.5–5.5 of sample accuracy would be detected. *The adaptation audit is weak:* 24–28 problems, 2–12 training examples per round that could teach the avoidance, and no positive control. *Cap-hit is reconstructed* from decoded text. *The verifier audit* used other models' solutions and inspected only rejections; the gold-storage defect was measured afterwards and ran unvaried through every run. *Every headline number is graded by the biased verifier* that also selects the data. *The* 512*-token cap binds* from the start, and the equal training limit, which includes the prompt, truncates a quarter of the pool. *Scope:* one model family (Qwen2.5) and one size in Design I, one verifier, one LoRA recipe with loss over the prompt, a restart-from-base design that leaves compounding little room, and possible test-set contamination. No arm improves on round 0, so we have no positive reference showing this recipe can help a 1.5B model. Appendix J gives the full list.

7 Conclusion

We asked whether generate–verify–select–train iterations make a 1.5B model better at mathematics or only teach it to write what a rule-based verifier accepts. On a held-out exam, the first answer is no gain that this design could detect: on pass@1 and pass@k, the pre-set accuracy measures, no run improves detectably,

though gains below about 10 and 8–10 points could have been missed; the intervals for sample accuracy (a primary measure chosen after the runs) exclude gains above about 2.2 points, a loose bound because they omit training-seed variance and are not an equivalence test. No arm improves, including training on reference solutions, so this is a null for the recipe. The second question stays open, because our pre-set rule for it could only fire if accuracy rose, and the post hoc `\boxed` share falls in a way that imitation of the less-boxed training pool and drift would both produce. A post hoc per-round false-reject audit, carried over from RLVR, finds no sign that the model learns to avoid one verifier defect once the falling `\boxed` share is accounted for, but with about 25 affected problems, a handful of training examples per round and no positive control it cannot rule such learning out. In a post hoc description, most clearly at the highest rate, the model becomes less accurate on the solutions it does finish; more generations hit the token cap only at the highest rate, and clearly only at one of its two seeds, where the change in cap-hit share accounts for 12–30% of the decline in the original seed-1 runs and 52–72% at seed 2; a character-length heuristic we first used overstated that part. Reference-solution training degrades the model more, with much shorter answers. Patching a rare verifier defect and removing the filter made no difference of consistent sign across measures and seeds (pass@1 flips sign between seeds; removing the filter lowers pass@4 at both seeds, within noise; single-seed intervals sometimes exclude zero), neither control could have detected a small effect, and a frequent defect in our gold storage, which rejects about one in six correct solutions, ran unvaried through every experiment. A sharp decline seen in an earlier pooled design does not reproduce under isolation and held-out scoring, and because that design also trained several times longer per round, we leave its cause open. We report the earlier readings that did not hold up because the designs that produced them are common.

A rising score, in self-training or in RLVR, does not by itself show that a model learned mathematics rather than its verifier. A per-round false-reject audit asks part of that question whether accuracy rises or falls, but our data show it must be conditioned on answer format and on the problems where the defect can be avoided, and it needs a positive control; applying it so to public RLVR checkpoints, where the pressure to adapt is direct, is a natural next step. Future work, including a dose-matched control for the gold-storage defect, is outlined in Appendix K.

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20. Zeng, W., Huang, Y., Liu, Q., Liu, W., He, K., Ma, Z., He, J.: SimpleRL-Zoo: Investigating and Taming Zero Reinforcement Learning for Open Base Models in the Wild. arXiv:2503.18892 (2025)

A Related Work in Full

Self-training and its limits. STaR [19] introduced the generate–filter–retrain loop, and ReST [12], ReST-EM [11] and RFT [17] formalise it; ReST-EM, which like us restarts fine-tuning from the base model each iteration, reports diminishing test gains after the first round and a one-round regression on APPS. ReST filters machine-translation samples with a learned reward model and a rising threshold; the others filter by answer correctness. None of them measures how the filter's errors depend on the form of the answer, and they mostly use larger models than ours. Yue et al. [18] show that RL raises pass@1 while the base model wins at large pass@k, and Teacher-Free Self-Training [13] observes a pass@8 versus pass@64 crossover on a free-verifier domain, concluding that self-training "amplifies but does not compound". Training repeatedly on model-generated data erodes distribution tails [10], and Qiao et al. [8] argue, in theory and with experiments, that a selector seeing only a narrow slice of the target distribution accelerates collapse.

Verifier error. TinyV [16] measures that rule-based verifiers reject many correct solutions and repairs them with an LLM check. From Accuracy to Robustness [6] reports an average recall of 86% for rule-based verifiers, i.e. about 14% of correct responses rejected, but for `math-verify` itself on their MATH subset a recall of 0.95 (about 5% false negatives; their Table 5, on DeepSeek-R1-Distill-Qwen-32B outputs). Our MATH-500 audit gives 7.8% (95% Wilson interval 4.3–13.8%) on another model's solutions, a figure whose relation to the loop's own rate is not known (Section B), and a pipeline-specific defect alone rejects 15.5–18.4% of the loop's own solutions that a repaired verifier accepts, concentrated on about a tenth of the problems. These figures differ in verifier version, solution source and denominator, so we compare them only in order of magnitude. Imperfect Verifiers [1] gives the most formal noise model, with fixed false-accept and false-reject probabilities, and notes that real noise can depend on formatting but then assumes content-independent rates. Egashira et al. [4] classify verifier errors in RLVR on a synthetic arithmetic task (Qwen3-1.7B-Base and OLMo3-7B, with Qwen2.5-1.5B-Instruct, our model, in an appendix) with hand-designed patterns such as format-based false negatives, and find that systematic false negatives behave like random noise whereas false positives (rewards given to answers that are near the truth, or that contain a keyword) can cause plateau or collapse; they also note that a domain whose queries are always rejected can at best plateau, which is the position of our unparseable-gold problems. They also track, per RL step, the false-negative rate on the policy's own rollouts and report that the policy first unlearns the behaviour that triggers false negatives; TinyV tracks the false-negative share of rollouts across GRPO steps (about a fifth per step in their runs). Our audit (Section 5.3) applies this measurement to offline SFT. Our setting differs in paradigm (offline rejection-sampling SFT with a fresh adapter per round, not on-policy RL) and in task (real benchmark problems). Our patch control cannot test their false-negative finding, because it changes too little of the training pool (Section F), and the much larger false-negative class we measured

afterwards is present in both arms. Nor does our no-filter control test their false-positive finding: their false positives are systematic (a fixed wrong pattern that is always accepted), whereas no-filter admits every sample, about two thirds of them incorrect in unstructured ways, under SFT with a 3–4× larger training dose. It does add false positives in the plain sense, wrong answers in the training pool, but unstructured ones, and 62.4% of its round-0 pool is also truncated in training (against 24.5% for Run A). It did not end consistently worse than the filtered run, which we report only as a related observation. Their finding that systematic false negatives act like noise would, if it carries over, predict that a model need not learn to avoid a false-negative pattern, which is consistent with our audit, but our audit has too little power to test it.

Auditing self-improvement. Xu et al. [15] audit three rounds of rank-32 LoRA self-training of Qwen3-8B ($\eta = 10^{-4}$; filters by answer match or majority vote, plus a distillation arm from an external teacher as a positive control) against a frozen untrained control, and identify seven measurement failures that each invert a reported finding when its control is absent, including training-seed variance and a fixed token cap under style drift: their distillation arm, the most effective one, scored as the most destructive because its truncation rate rose from 16.5% to 65.3%. Self-training does not improve rarely-solved problems and corrupts problems solved at baseline. This is the study closest to ours in spirit. Our per-problem added/lost counts point the same way (more problems lost than gained), though without their frozen control and with net counts that only restate the change in pass@k (Section 3.4), and our adapters also drift in style (longer outputs, fewer `\boxed` answers), and a fixed token cap matters in our setting too: the untrained model already hits it in 41–48% of MATH-500 generations. Their finding is a warning that a decline under a fixed cap can be misread as lost capability; this is why we separate finished accuracy from termination and re-score at a larger cap (Section H). Their setup differs in model size, rank and filters; we lack their per-round frozen arm (we re-score the untrained model once, Section H) and a distillation arm, and we add a measured format-biased verifier, a learning-rate series and reference-solution and no-filter controls.

Reference-guided training. Kim et al. [7] show that self-distillation conditioned on the ground-truth solution can cut mathematical accuracy by up to 40% (Qwen3-1.7B/8B, DeepSeek-Distill-Qwen-7B and Olmo3-7B-Instruct) while *shortening* responses, which they trace to suppressed epistemic verbalization. Our train-on-gold control is plain SFT on human reference solutions, not self-distillation, so it is at most a loose analogue; it shows a similar surface signature (shorter responses, lower accuracy) that may simply come from imitating terse references. We did not measure epistemic markers.

Positioning. In the literature we surveyed (searched through October 2026) we found no multi-round self-training study that uses a rule-based verifier with a measured format bias together with a matched patched-verifier control; the closest are Egashira et al. and Xu et al. Given that the patch turned out to

be too weak to test anything (Section F), we do not present this combination as the contribution. The contribution is the absence of a detectable gain on the pre-set accuracy measures, with intervals and the smallest detectable effects, the description of what training at higher rates changes, and the design lessons. The per-round false-reject audit of Section 5.3 is not new: Egashira et al. track the false-negative rate on the policy's own rollouts per RL step and TinyV the false-negative share across GRPO steps. What we add is its application to offline SFT self-training, where a false reject only removes an example, and the conditions it needs there (conditioning on answer format and on the problems the defect affects, and a count of how many training examples could teach the model to avoid the defect).

B The Verifier in Full

Format probe. We re-rendered gold answers from GSM8K and MATH-500 in individually valid formats (bare number, trailing-zero decimal, `\frac`, `\dfrac`, `\tfrac`, boxed, dollar-delimited, sentence, with units) and asked the verifier to grade them against the original gold. On GSM8K plain rewrites are never rejected, while `\dfrac`/`\tfrac` are rejected 200/200 (the command is not parsed). On MATH-500, `\dfrac`/`\tfrac` are rejected 146/146 (100%), trailing-zero decimals 13/22, and boxed forms 41/200. Of those 41, 24 are gold answers our pipeline stores in forms the verifier cannot parse at all, so part of this figure measures our gold storage, not the verifier. The other 17 (17 of the 200 boxed probes, 8.5%) are compound answers such as $3\sqrt{13}$: the gold is stored without a delimiter, and `math-verify` applies its full LaTeX extraction only inside `\boxed{}` or `$...$`, so the bare gold is parsed to its first number and no longer matches a correct boxed prediction.

False-negative rate on real solutions. We took one public solution per problem for the first 200 problems of each dataset (teacher traces for GSM8K, Qwen2.5-7B-Instruct rollouts for MATH-500), not the 1.5B model's own outputs. Correctness was established by a scripted comparison of the boxed answer with the gold string after normalisation (`src/dacntt/read.py`: delimiters, braces, spaces and commas removed, `\dfrac`/`\tfrac` read as `\frac`, simple fractions compared by value), and we then counted how many of these correct solutions the verifier rejects. The false-negative rate is 7.8% on MATH-500 (10 of 128 correct solutions; 24 problems with unparseable gold excluded) and 0% on GSM8K (0/200; 95% Wilson interval 0–1.9%, one-sided upper bound 1.3%), identical for the `default` and `reward` extraction presets of `math-verify`. Of the ten MATH-500 cases, eight are rejected when the whole solution is verified but accepted when only the boxed answer is, which we labelled wrong-token extraction from a long solution, and two are rewrites the normalisation equates but the verifier does not match (e.g. `6 + 9i` for `6+9i`). Most of the eight are compound answers ($3\sqrt{13}$, $6 - 5i$, $11\sqrt{2}$, a list, a polynomial), so they overlap with the compound class of the format probe and very likely with the gold-storage defect below (verifying the boxed answer alone compares two bare strings, which the bare-gold parse

matches); the label describes where the failure shows, not its cause. The 95% Wilson interval for 10/128 is 4.3–13.8%. The 7.8% estimates the false-negative rate on Qwen2.5-7B-Instruct solutions for problems whose gold the verifier can parse, among answers our normalisation recognises as correct. Correct answers in forms it does not equate are left out of the denominator, and the 24 problems with unparseable gold are counted as a separate defect below; relative to the loop's own rate, where the 1.5B model writes longer and, after training, less boxed solutions, the direction of the difference is not known. The audit also inspected only rejected solutions, so it cannot measure the false-positive rate; that the verifier rarely accepts wrong answers is an assumption here, not a measurement.

Unparseable gold in the loop. Over all 500 MATH-500 problems the verifier cannot parse 51 stored gold answers (10.2%); 11–18 of them fall in each 150-problem exam partition. The loop keeps these problems: every sample is rejected, so they never enter the training pool and always count as unsolved. They lower absolute pass rates by up to 12 points but are constant within a partition, so they cancel in within-run and same-partition differences; they do not cancel in ratios or in levels compared across partitions. Wrapping the gold in `\boxed{}`, the re-score described below, makes 46 of these 51 golds parseable, so the two corrections overlap and should not be added (`experiments/viec3/review_cpu/dot4/unparsed.py`).

The patch. For the control we repaired the most *severe* identified failure mode with a normalisation pass that rewrites `\dfrac` and `\tfrac` to `\frac` before parsing, removing the 100%-rejection class and nothing else. The patched verifier is therefore still biased: it keeps the eight audited "wrong-token" false negatives (most of them probably the gold-storage defect) and the gold-storage defect itself.

The gold-storage defect in the loop's own outputs (post hoc). After the runs we re-scored the saved generations with a second normalisation that wraps the gold in `\boxed{}` when the prediction contains one and the gold does not (`experiments/viec3/review_cpu/`: `patch_dose.py`, `dose.py`, `flips.py`). This defect is far more frequent than the `\dfrac` class. Over all 500 problems it rejects 15.5–18.4% of the generations the re-scored verifier accepts (Run A at 5×10^{-4}, seeds 1 and 2, and at 10^{-5}, seed 0, every round). The kept pool covers only the 350 selection problems, where the share is about the same (17.8–18.8% at round 0): accepting these generations would enlarge the round-0 pool by about a fifth (Run A at seed 1: 440 → 542, +23%; at 10^{-5}: 872 → 1,066, +22%), against 0–4 examples (under 1%) for the `\dfrac` patch. In a hand check of 30 random flips, 29 pairs were the same value written differently (e.g. `\frac43` and `\frac{4}{3}`; one prediction had an empty box). The 15.5–18.4% assumes that the re-scored verifier itself accepts no wrong answer, which we did not measure. On the exam set the defect lowers absolute pass@1 by 6.7–8.0 points at round 0 and throughout the 10^{-5} runs, by 4.7 points at round 2 under 5×10^{-4}, and by 0.7–3.3 points for train-on-gold after round 0. The absolute offset is therefore not constant. As a share of re-scored exam pass@1 it is 6.7/40.0 and 4.7/28.0 (both

17%) for Run A at seed 1, 6.7/38.0 = 17.6% and 4.7/32.0 = 14.7% at seed 2, and about 20% throughout at 10^{-5}. Over all 500 problems and all samples (the r_t of Section 5.3, a different denominator) it falls from 18.3 to 15.7% and from 18.1 to 15.5% at 5×10^{-4} and from 18.3 to 16.8% at 10^{-5}; Section 5.3 shows that this fall follows the falling `\boxed` share and that the defect stays at 94–98% on the problems where it fires. With the original verifier the declines below are *understated*: the original-verifier sample-accuracy changes are 0.73–0.78 of the re-scored ones in the three self-training runs and 0.80–0.85 in train-on-gold. These ratios are observations for five runs, not a correction factor: intervals do not scale with them, so we do not rescale the other runs or the 2.2-point bound. We keep the original verifier's numbers in every table, since that is the verifier the loop trained with; they are therefore 4.7–8.0 points too low in absolute terms for self-training. Re-scored, with paired bootstrap intervals as in Section 3.4, sample accuracy changes by −10.7 [−14.8, −6.7] and −6.0 [−9.7, −2.2] points for Run A at 5×10^{-4} (seeds 1 and 2; original verifier −8.3 and −4.7), by −3.3 [−5.9, −0.7] for Run A at 10^{-5}, seed 0 (original −2.4), and by −24.7 and −22.0 for train-on-gold. In pass@1 the same runs go from 40.0% to 28.0% and from 38.0% to 32.0% (interval $[-14.0, +2.0]$ for the latter), from 40.0% to 36.7%, and from 40.0% to 16.0% and from 38.0% to 13.3%. Only these five runs were re-scored, three of them self-training; that the other runs' declines would also survive a re-score is an extrapolation from these three. Both arms of the patch control (Section F) carry this defect.

C Uncertainty in Full

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap: we resample the 150 exam problems with replacement (10,000 times), recompute the statistic for both rounds or runs on the same resampled problems, and take the 2.5th and 97.5th percentiles. Resampling problems rather than samples respects the clustering of the k samples within a problem. The per-sample correctness was recomputed from the saved generations with the same verifier; it reproduces the reported pass rates except for six round values that differ by one problem (0.6–0.7 points), most likely because of `math-verify` comparison timeouts (`experiments/viec3/bootstrap_ci.py`). The post hoc measures of Section H (Table 12, the matched column of Table 5 and the decomposition) use a separate re-scoring of the same generations (`review_cpu/scored/`), which differs from the values behind Tables 4 and 5 by up to one problem in some further rounds (e.g. 10^{-4}, seed 1, round 2: pass@1 33.3 against 34.0%); no conclusion depends on these differences. These intervals cover the choice of problems and the sampling of generations for a given trained adapter; they do *not* cover training-seed variance, which we can only see informally across our two or three seeds, so they understate the uncertainty of run-level conclusions. Our own data suggest by how much: a replicate of Run A at 5×10^{-4}, seed 1, with the same configuration,

Table 9. The two designs. They differ in more than the intended variables; the most quantifiable unintended difference is the training dose per round (about 4.5× as many optimizer steps in Design P).

	Design P (pooled)	Design I (isolated)
Datasets per run	GSM8K + MATH-500	one only
Adapter	one, shared	one per dataset
Training mixture	≈ 69% GSM8K	single-domain
Evaluation items	same as selection	held-out 30%
Training examples, round 0 (1.5B)	3,962	872 (A), 1,901 (C)
Verifier control	none	Run B (patched)
Models	0.5B and 1.5B	1.5B

ends 3.2 points of sample accuracy away from the original (Section H); and at seed 2 Runs A and B, whose round-0 training pools are identical (424 examples each), end 7.3 points apart in round-2 pass@1 (interval $[-14.0, -0.7]$, Section G), though that is a single interval near zero. A single interval that excludes zero is therefore not sufficient evidence of a reproducible effect, and one that includes zero does not rule an effect out. We also make no correction for multiplicity: Table 5 alone holds 76 intervals and the text about ten more. If no change were real, about four of the 76 would exclude zero by chance; many changes in the table are clearly real (the train-on-gold and GSM8K declines), and the intervals are not independent, so this is not a test that any particular interval is a false alarm, only a reason to rest conclusions on patterns across runs and seeds rather than on single intervals. Our only estimate of training-seed variance rests on one same-configuration pair, so it is itself very imprecise. Single proportions use Wilson intervals.

D Design P in Full

Table 2 reports pass@1 and pass@8 for both models over five rounds of the pooled design, as computed during the run. A later recomputation (Table 10) reproduces every pooled value with both verifier presets except round-3 pass@8, where Table 2's 63.6% matches the `default` preset and the `reward` preset used for selection gives 63.5%. The 1.5B decline is severe and accelerating; the 0.5B decline is small (−3.7 and −5.6 points). Splitting the 1.5B run by dataset (Table 10) shows both datasets falling heavily, although GSM8K has a measured 0% false-negative rate, though that rate was measured on external solutions in the untrained model's style, not on Design P's late-round outputs. The accepted pool also contracts every round (1.5B: 3,962, 3,730, 3,395, 2,638, 1,566; 0.5B: 2,533 to 2,123; `experiments/viec3/review_cpu/dot4/poolP.py`) and the 1.5B output style shifts, with `\boxed{}` falling from 5,608 to 2,693 of 8,000 generations while dollar-delimited math rises from 1,453 to 3,406.

Design P cannot be interpreted causally, for three reasons. First, each round trains one adapter on the union of both datasets' accepted solutions, of which

Table 10. Design P, 1.5B, split by dataset, recomputed from the saved generations (preset `reward`, the one used for selection; the `default` preset differs only at MATH-500 round-3 pass@8, 45.0%). Pooled, the recomputed round-3 pass@8 is 63.5% (`reward`) and 63.6% (`default`), against 63.6% in Table 2.

Dataset	Metric	R0	R1	R2	R3	R4	Δ
GSM8K	pass@1	69.6%	62.0%	61.0%	45.6%	26.6%	−43.0 pt
GSM8K	pass@8	91.4%	90.4%	89.0%	82.2%	65.8%	−25.6 pt
MATH-500	pass@1	30.4%	30.4%	28.0%	19.2%	10.6%	−19.8 pt
MATH-500	pass@8	53.8%	52.6%	51.4%	44.8%	31.4%	−22.4 pt

GSM8K supplies 68.4–70.2% in every round, so either dataset's trajectory is a joint function of both pools. Second, it scores each round on the problems it also used to select training data. We expected this to *raise* later-round scores, if anything. Design I lets us check that expectation, since it also generates on its 350 selection problems: there, Run A's sample accuracy on the selection problems falls as much as or more than on the exam at the two higher rates (e.g. Run A at 5×10^{-4}: −8.2 and −5.3 points against −8.3 and −4.7 on the exam; at 10^{-4}, seed 1: −6.2 against −1.5; no intervals); at 10^{-5} it falls slightly less (−1.0 and −1.1 against −2.4 and −1.5), and Run C's selection problems fall less at 5×10^{-4} (−11.3 against −15.3) and slightly more at 10^{-5} (−13.8 against −13.1; exam values from Table 5). Training on a problem's own accepted solutions therefore did not protect it, so selection-set scoring is unlikely to have masked a decline; since the selection problems sometimes fall more, it may have enlarged one, and it remains a candidate contributor. Third, and not anticipated when we designed the follow-up, Design P trains on far more examples per round: 3,962 at round 0 for the 1.5B model, against 872 for the isolated MATH-500 run at the same learning rate. With one epoch and the same batch size this is about 4.5× as many optimizer steps (Design P's training logs were not saved, so this is inferred from pool size). Section G suggests, on the evidence of one seed, that a higher learning rate produces more decline, with a falling `\boxed` share as here (pass@1 falls faster than pass@k in only two of its four runs at 5×10^{-4}), and Run C, which takes about twice Run A's steps per round at the same rate, is the one 10^{-5} run that declines clearly. One reading consistent with these data is that Design P's adapter changed the model more. It rests on two assumptions we have not tested: that more optimizer steps act like a higher learning rate, and that Run C's decline comes from its dose rather than its dataset (a single comparator). The time course adds a constraint: with a fresh adapter each round, the round-t model is trained on the round-$(t-1)$ pool alone, so the restart-from-base design leaves the data as the only channel from one round to the next. The pools shrink, yet the largest pass@1 drops (−12.1 and −13.8 points) come from the two smallest pools (3,395 and 2,638), so dose per round cannot by itself explain the acceleration. Dose and data need not be alternatives, though: a large dose in an early round can change what the model generates, and so what the next pool contains, which is one way dose could act

through the data. The 0.5B model, trained on 2,533 examples at round 0 (about 2.9× Run A), declines only a little (−3.7 points of pass@1), which weighs against dose alone, though it is a different model with a smaller pool. Pooling or scoring may also have contributed; no run separates these explanations.

E Design I at 10^{-5} in Full

Table 3 reports the isolated runs on their held-out exam sets. pass@8 ends within 4 points of its round-0 value in every run. On MATH-500, pass@1 is 2.7–5.3 points lower at round 1 than at round 0 and roughly flat afterwards, and 2.7–4.0 points lower at round 4; the bootstrap intervals for these round-0-to-round-4 changes are between $[-10.7, +2.7]$ and $[-9.3, +4.0]$, all including zero (Table 5). With one first-sample draw per problem per round, a step at round 1 cannot be distinguished from noise around a flat or slightly negative trend, so we claim only that there is no evidence of compounding on MATH-500 at this rate (Run C, below, loses most of its pass@1 in its last round), not that the first round has a one-off cost. The less noisy measures do move: sample accuracy falls by 1.2–3.2 points in these four runs, with one interval (B, seed 0) excluding zero, and the (post hoc) matched finished accuracy falls by 1.1–4.2 points, with intervals excluding zero at seed 0 for both A and B. These seed-0 intervals only just exclude zero (upper ends −0.6 and 0.0 for sample accuracy, −0.5 and −0.9 matched), A and B at seed 0 share their round-0 generations and almost all of their first pool, so they are close to one replicate, and at seed 1 nothing excludes zero. Because these are near-replicate intervals that only just exclude zero, we read this as a possible small decline at this rate, not an established one; what we do not see is a gain of the size the design could detect. Run C (GSM8K) declines more: pass@1 falls 12.7 points, 9.3 of them in the final round (63.3% → 54.0%; interval for the whole change $[-21.3, -4.0]$), and sample accuracy 13.1 points, while pass@8 falls 2.7 points (interval including zero) and the cap-hit share rises only from 3.5% to 4.2%. Finished accuracy falls 12.8 points matched, and the decomposition of Section H puts 97% of the sample-accuracy decline in finished accuracy, so this decline is not a termination effect. It is a single run, and it took about twice as many optimizer steps per round as Run A (951 against 436 at round 0), so we do not attribute it to the dataset alone.

How much training happened at 10^{-5}. The training loss does fall at this rate: within round 0 of Run A it goes from 0.72 to 0.38 (seed 0) and from 0.70 to 0.40 (seed 1) over about 420 steps, with an epoch mean of 0.47, against epoch means of 0.24–0.30 at 10^{-4}–5×10^{-4}. The adapter therefore learns, but slowly, and it changes the outputs little: mean length and the share of generations that hit the token cap are essentially unchanged from round 0 to round 4 (cap-hit 46.9 → 47.4% at seed 0 and 40.7 → 41.8% at seed 1 for Run A; Table 12). This holds for the MATH-500 runs at about 420–436 steps per round; Run C at the same rate, with about twice the steps and a different dataset, does change its outputs. With that one comparison we cannot tell whether "little change"

depends on the steps, the dataset or both. Length and cap-hit share are coarse statistics, and we did not measure finer ones (e.g. per-token divergence from the base model), so the evidence that the adapter did little is weak. A flat pass@k curve in the MATH-500 runs is therefore not evidence that GVT is stable.

Noise. Round 0 is the untrained base model, so round-0 differences between seeds mostly measure how hard each partition is: round-0 pass@1 is 32.0% at seed 0 and 36.7% at seed 1, and on the third partition used later round-0 pass@4 is 40.0% against 50.0% and 51.3% on the other two (Table 4). These levels cancel in within-run changes. Noise in the changes themselves is better seen elsewhere: re-sampling the same model on the same problems moves pass@1 by up to 6.7 points (Section H), and a single 150-problem pass@1 has a standard error of about 3.8 points. Agreement across seeds tests partition and training variance together; training variance alone is seen only in same-partition pairs (A against B at the same seed, Section G, and the replicate of Section H). We therefore report the bootstrap intervals above rather than judge differences against one "noise floor".

F Patch Control and Answer Diversity in Full

Runs A and B are identical in dataset, seed, split, model and training configuration and differ only in whether `\dfrac`/`\tfrac` are normalised before verification. Their curves are close (Table 3): at round 4, B minus A is -0.7 [-3.3, $+2.0$] points of pass@1 at seed 0 and $+0.0$ [-2.7, $+2.7$] at seed 1 (95% intervals). Run B's exam is graded with the patched verifier, so these differences also contain a grading offset: at seed 1 the identical round-0 generations already score 0.6 points higher in B, so the model part of the seed-1 difference is about -0.6. Narrow intervals here bound the difference between two nearly identical trajectories, not a patch effect. This null is uninformative, for two reasons we should have checked before running it. First, the treatment is tiny: `\dfrac` appears in 20–29 of the 4,000 generations per round (0.50–0.73%), `\tfrac` never, and only the correct ones among those are wrongly discarded; the kept pools of A and B differ by 4 examples at seed 0 (872 vs 876) and 3 at seed 1 (833 vs 836), under 0.5% of the pool. Second, at 10^{-5} and the step count of these MATH-500 runs the adapter changes the outputs little (Section E), so two nearly identical pools would give nearly identical models whatever the verifier did. We cannot separate these two explanations. The result is therefore: *removing one rare format defect had no detectable effect*, which the prevalence count alone predicts; it is not evidence about directional verifier bias in general, and it leaves untouched the gold-storage defect, which rejects 15.5–18.4% of correct in-loop solutions in both arms (Section B). Verifier-bias figures in the literature, such as the $\sim$14% of [6], are likewise conditional on a reference distribution; what matters for a loop is how often the affected forms occur in that loop's own output, which is cheap to count.

Pre-filter answer diversity. Over all problems of each run (in Design I the 500 selection and exam items together, not the exam alone) we also measured, before filtering, the fraction of problems on which all $k = 8$ samples give the same extracted answer, for Design P's 1.5B run and the 10^{-5} Design I runs at seed 0. It falls in every one: 24.8% → 1.7% for Design P's 1.5B run, 33.6% → 16.0% for Run C, 14.6% → 12.0% (A) and 14.6% → 12.2% (B). This agreement is mechanically tied to accuracy (problems on which every sample is correct are all-agreement problems) and to generations that hit the token cap, so it adds little beyond the accuracy change and does not by itself rule out stylistic convergence, which concerns how answers are written rather than which answer is given. The fraction of distinct answers among the k samples, which we had planned as the primary diversity measure, barely moves at 10^{-5} in the MATH-500 runs (0.542 → 0.554 for Run A, seed 0; 0.528 → 0.543 for Run B, seed 1) but rises for Run C (0.359 → 0.500) and in Design P (0.443 → 0.712). We report both for completeness and draw no conclusion from them.

G Higher Learning Rates in Full

Because 10^{-5} changed the MATH-500 outputs little, we reran the MATH-500 runs at higher rates with a smaller budget (three rounds, $k = 4$, so pass@4 replaces pass@8 and the pools are about half as large): Runs A and B at 5×10^{-4} at two seeds, and Run A at 10^{-4} at three seeds and at 2×10^{-4} at two. Seed 1 shares the exam partition of the seed-1 runs above and seeds 2 and 3 draw new ones. These runs are not directly comparable with the $k = 8$, five-round runs in metric or training dose. Within round 0 the loss falls from about 0.5 to 0.25 at 5×10^{-4} and from about 0.6 to 0.28 at 10^{-4} over about 220 steps.

At 5×10^{-4} every run declines, with wide intervals. By round 2 pass@1 has fallen 10.0 and 4.0 points (A) and 4.7 and 11.3 points (B), and pass@4 7.3, 6.7, 2.0 and 11.3 points (Table 11). The bootstrap intervals (Table 5) exclude zero for two of the four pass@1 changes and one (two more reach zero at their upper end) of the four pass@4 changes, and for the sample accuracy of all four. A and B at the same seed share the partition, the round-0 generations and almost the whole first pool, so the four runs are closer to two independent replicates than four. We read this as a consistent but imprecisely measured decline. The style shift does not separate imitation from drift. The verifier-accepted pool is 89–91% `\boxed` at round 0, far more than all generations (52–54%, Table 11), because unfinished generations are rarely boxed and rarely accepted. But the right comparison is with *finished* generations, and these start *more* boxed than the pool (95.6% and 98.5% on the exam at seeds 1 and 2; the pool also holds truncated and cap-hit examples that lose their final `\boxed`). Among finished exam generations the share falls to 69.7% (A, seed 1) and 82.9% (A, seed 2), from above the round-0 pool to below it, while the pool, drawn from these generations, falls too (73–77% by round 2). A model imitating its less-boxed training data would move the same way, so this share cannot separate imitation of the pool from drift.

Table 11. Runs A and B at 5×10^{-4} (MATH-500, $k = 4$, three rounds), held-out exam set of 150 problems. B differs from A only in the verifier patch; each seed draws its own exam partition. "Kept pool" is the number of training examples (= 2× optimizer steps). "Boxed" is the share of the 2,000 generations in each round containing `\boxed`.

Run, seed	Round	pass@1	pass@4	Kept pool	Boxed
A (orig.), 1	0	33.3%	50.0%	440	52%
	1	32.7%	49.3%	391	51%
	2	23.3%	42.7%	326	36%
B (patched), 1	0	33.3%	50.7%	443	52%
	1	35.3%	50.0%	392	49%
	2	28.7%	48.7%	328	41%
A (orig.), 2	0	31.3%	51.3%	424	54%
	1	31.3%	46.0%	386	50%
	2	27.3%	44.7%	350	37%
B (patched), 2	0	31.3%	52.0%	424	54%
	1	31.3%	44.7%	398	50%
	2	20.0%	40.7%	332	38%

The decline may depend on the learning rate, on the evidence of one seed. Only seeds 1 and 2 were run at all three higher rates. At seed 1, sample accuracy at round 2 falls by 1.5, 3.5 and 8.3 points at 10^{-4}, 2×10^{-4} and 5×10^{-4}; at seed 2 by 3.7, 3.8 and 4.7, which is almost flat (Table 5; seed 3, run only at 10^{-4}, gives -1.0). The seed-1 series is monotone, the seed-2 series barely so, and at 10^{-4} and 2×10^{-4} the individual changes are about the size of re-sampling noise, so the rate dependence is suggestive, not established. The share of generations that hit the token cap does *not* follow the rate: it is flat or falls at 10^{-4} and 2×10^{-4} and rises only at 5×10^{-4} (Table 4).

The patch has no consistent effect. At round 2, B minus A is +5.3 [−2.0, +12.0] points of pass@1 at seed 1 and −7.3 [−14.0, −0.7] at seed 2. The sign flips with the seed, and the seed-2 interval excludes zero although the patch changes the first training pool by 0–3 examples. Same-seed runs that start from almost the same pool can thus end several points apart once the adapter moves (their later pools differ by 1–18 examples as the trajectories separate). This is one pass@1 interval whose upper end only just clears zero, so on its own it is weak evidence of trajectory noise; the replicate of Run A at seed 1 (Section H), which ends 3.2 points of sample accuracy and 4.1 points of matched finished accuracy away from the original, points the same way.

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem regardless of correctness (1,400 examples per round), removing the verifier from the loop. The *train-on-gold* run ignores the model's generations and trains on the 350 reference solutions, identical every round. Both were run at two seeds, on the partitions of Run A at the same seeds. Neither is a single-variable control: per round, no-filter takes 700 optimizer steps and train-on-gold

175, against 220, 196 and 163 for Run A at seed 1 (and 212, 193, 175 at seed 2). Train-on-gold is thus close to Run A in optimizer steps, so its comparison is mainly one of data; no-filter takes 3–4× as many steps, and since dose may matter (the rate dependence above is suggestive), dose and data differ at once in that comparison.

Removing the filter is never better than keeping it on any measure beyond noise, and is worse on pass@4 at both seeds, but the dose difference means this cannot be attributed to the filter. At round 2, no-filter minus Run A is +4.0 [−2.7, +10.7] (seed 1) and −4.7 [−12.0, +2.7] (seed 2) points of pass@1, of mixed sign, and −2.0 [−8.7, +4.7] and −5.3 [−12.0, +1.3] points of pass@4, negative at both seeds, so pass@4 weakly favours keeping the filter; for sample accuracy the difference is +0.2 [−3.3, +3.7] at seed 1 and −5.5 [−9.2, −1.8] at seed 2, so one single-seed interval excludes zero, in favour of the filter, and the other is centred on zero. The verifier gate keeps 30–31% of the samples at round 0 and 23–25% by round 2. With two seeds, a 3–4× larger training dose for no-filter, and intervals of this width, we can only say that effects of a few points in either direction are not excluded. Training on reference solutions is followed by a much larger decline: pass@1 falls to 18.0% and 15.3% (seed 1) and 8.7% and 10.0% (seed 2), and pass@4 by about 21 points by round 2 at both seeds ($[-29.3, -12.7]$ at worst), while the model adopts the reference style (generations with a `$`-delimited span go from 5% to 98%, with `\boxed` from 52% to 86%). That accuracy is graded by the same verifier after a large format shift, so part of this drop could be verifier error. We checked this in two ways after the runs (post hoc). First, among the round-2 answers the verifier rejects, those whose final `\boxed` content equals the stored gold string after light normalisation are 14 of 600 generations at seed 1 and 11 of 600 at seed 2; counting them correct would raise round-2 sample accuracy by about 2 points. This is a lower bound: exact string match misses equivalent forms, and the about 14% of round-2 generations with no `\boxed` at all are not covered. Second, the gold-storage re-score of Section B makes the sample-accuracy decline larger, not smaller (−24.7 and −22.0 points against −19.7 and −18.7), because the defect penalises round 0 more; but it repairs one defect and does not test whether answers in the new style are parsed correctly. Neither check is a hand audit. Even several times the first bound would be far below a fall of about 19 points, and finished accuracy falls to 12–15%, so we think the fall is unlikely to be mainly verifier error. But both checks bound verifier error only from below, the unboxed generations are not checked at all, and we have no upper bound, so a format-shift error of unknown size remains. Its rounds are replicate fine-tunes on identical data (their final training losses agree to three decimals), so the difference between its rounds 1 and 2 mainly reflects generation sampling. No verifier-filtered self-training run at this rate shows a pass@k gain; their pass@4 falls by 2.0–11.3 points (no-filter 9.3 and 12.0), whereas train-on-gold falls by about 21.

Run C at 5×10^{-4}. We also reran Run C (GSM8K, seed 1, three rounds, $k = 4$) at 5×10^{-4}, once. pass@1 falls 70.7% → 64.7% → 48.7% (interval for the round-0-to-2 change $[-29.3, -15.3]$) and pass@4 86.0% → 86.0% → 76.7%. The cap-hit

share rises from 3.8% to 15.0% and mean length from 958 to 1,313 characters, and accuracy on the finished generations falls clearly (71.9% → 63.3%; −12.1 points matched). In the three-part split of Section H, about half of this run's sample-accuracy decline (51%) is in the finished-accuracy part and 43% in the cap-share part (Table 7), so on GSM8K, in this single run, the loss is both less accurate solutions and more unfinished ones. This run uses a different partition, k and round count from the 10^{-5} Run C, so the two can be compared only through their within-run changes (−1.3 points of pass@1 by round 2 at 10^{-5}, −22.0 at 5×10^{-4}; the 10^{-5} run falls further later, to −12.7 by round 4). That is suggestive of a rate dependence on GSM8K too, from one run per rate.

H Anatomy of the Decline in Full

Everything in this subsection is post hoc: the cap-hit measure, the matched metric and the decomposition were defined after the runs, to explain the declines, and they describe the declines rather than identify causes. Sample-level accuracy mixes two things: whether a model that finishes a solution gets it right, and whether it finishes within the 512-token cap at all. Token counts were not saved at generation time, so we re-tokenised the saved decoded text with the model's tokenizer and call a generation *cap-hit* if it has at least 508 tokens (`experiments/viec3/review_cpu/caphit.py`, `capanat.py`); the margin allows for small differences between generated and re-encoded token sequences. The threshold matters little: few generations fall near it (at most 37 of 600 per run and round have 480–507 tokens), a threshold of 500 raises the cap-hit share by at most about 2 points and one of 512 lowers it by at most about 4.0, and with either the share still rises only at 5×10^{-4}. The untrained model already hits the cap in 40.7–48.2% of its MATH-500 exam generations, depending on partition and sampling (43.0–43.2% on the $k = 4$ seed-1 and seed-2 partitions; 3.5–3.8% on GSM8K), and only 4–7% of those are accepted. The 512-token cap is therefore binding from the start, not a consequence of training.

An earlier proxy overstated termination. Earlier versions of this paper called a generation *degenerate* if it was longer than 1,500 characters and contained no `\boxed`, and read its rise (9.8% → 41.3% for Run A, seed 1) as a growing failure to finish. Measured against cap-hit, the proxy flags only 9.8% of round-0 generations when 43.2% hit the cap, and its rise mostly tracks a change in the text rather than in termination: proxy-flagged generations have 3.3 characters per token at round 0 and 4.9–5.0 at round 2, against 2.7–3.0 for the rest, so drifting prose crosses 1,500 characters within the same token budget. At round 0 every proxy-flagged generation is a cap-hit (at round 2, 80–98% are), but the proxy catches only 19–25% of cap-hits at round 0 and 58–75% at round 2 (at 5×10^{-4}), so its rise is largely a rise in the share of cap-hits it can see. At 10^{-4} and 2×10^{-4} the proxy share grows 2.0–2.6× (at seed 1, 9.8% → 19.5% at 10^{-4} and 9.8% → 25.0% at 2×10^{-4}) while cap-hit stays flat or falls.

At the highest rate, finished solutions are less accurate. In all six self-training runs at 5×10^{-4} (Table 12) mean length grows by 41–52%, but the cap-hit

Table 12. What the decline consists of (MATH-500, exam generations, round 0 → last round). "Cap-hit" is the share of generations whose decoded text re-tokenises to ≥ 508 tokens; "finished acc." is sample accuracy on the other generations, unadjusted; "matched Δ" is last-round finished accuracy minus the untrained model's finished accuracy on the same problem mix (intervals in Table 5). "Trunc." is the percentage of the round-0 training pool longer than the 512-token training limit. The first six rows are at 5×10^{-4}, then 2×10^{-4} and 10^{-4}; the 10^{-5} rows are the original runs ($k = 8$, round 0 → round 4, not comparable in dose); the last two rows (train on gold) are at 5×10^{-4}.

Run	Seed	Length (chars)	Cap-hit (%)	Fin. acc. (%)	Matched Δ	Trunc.
A (orig.)	1	1175 → 1678	43.2 → 46.2	54.0 → 43.0	−9.1	24.5
B (patched)	1	1175 → 1659	43.2 → 47.8	54.0 → 46.3	−6.8	24.6
A (orig.)	2	1154 → 1641	43.0 → 51.2	50.6 → 45.4	−5.5	24.5
B (patched)	2	1154 → 1728	43.0 → 54.0	51.2 → 44.2	−5.9	24.5
No filter	1	1175 → 1668	43.2 → 45.5	54.0 → 43.1	−10.6	62.4
No filter	2	1154 → 1755	43.0 → 57.2	50.6 → 43.6	−6.6	—
A, 2×10^{-4}	1	1175 → 1433	43.2 → 40.8	54.0 → 46.5	−4.0	24.5
A, 2×10^{-4}	2	1154 → 1431	43.0 → 39.7	50.6 → 41.7	−4.7	24.5
A, 10^{-4}	1	1175 → 1276	43.2 → 42.7	54.0 → 48.5	−4.3	24.5
A, 10^{-4}	2	1154 → 1284	43.0 → 42.8	50.6 → 44.6	−3.5	24.5
A, 10^{-4}	3	1178 → 1355	48.2 → 45.7	50.2 → 47.5	−0.6	—
A, 10^{-5}	0	1158 → 1147	46.9 → 47.4	54.8 → 51.0	−3.8	—
A, 10^{-5}	1	1158 → 1146	40.7 → 41.8	55.2 → 54.3	−2.3	—
Train on gold	1	1175 → 485	43.2 → 7.2	54.0 → 14.7	−25.6	17.4
Train on gold	2	1154 → 516	43.0 → 8.3	50.6 → 12.5	−25.9	—

share rises by only 2.3–14.2 points, while accuracy on the generations that finish falls from 50.6–54.0% to 43.0–46.3%. Because the problems that still finish change, we compared last-round finished accuracy with the untrained model's finished accuracy on the *same problem mix*, weighting each problem's round-0 finished accuracy by the number of finished last-round samples it has. Problems with no finished last-round sample drop out (22–45 of the 150 per run); where a problem had no finished round-0 sample we use its accuracy over all round-0 samples (6–21 problems, 34–37 in the gold runs), which, being mostly cap-hit accuracy, biases the baseline down for those problems; intervals use 2,000 problem-level bootstrap resamples. Because training changes which problems finish, the matched metric conditions on an outcome of training and is descriptive, not a causal effect. On that matched basis finished generations lose 5.5–10.6 points, with intervals excluding zero in five of the six runs (Table 5). We also split the change in sample accuracy symmetrically into a finished-accuracy part $\bar{s}\,\Delta f$, a cap-share part $\Delta s\,(\bar{f} - \bar{c})$ and a cap-accuracy part $(1 - \bar{s})\,\Delta c$, where s is the share of finished generations, f and c the accuracy of finished and cap-hit generations, and bars denote round-0/last-round means. Only the cap-share part measures termination; the cap-accuracy part is a change in how often unfinished text is still accepted. Table 7 gives the three parts and paired

bootstrap intervals (2,000 resamples) for the cap-hit change and for the finished-accuracy share (`experiments/viec3/review_cpu/dot4/boot.py`). This decomposition uses the unmatched finished accuracy, so its finished part still contains the shift in which problems finish.

At 5×10^{-4} the cap-hit rise has an interval excluding zero in all three seed-2 runs and in B at seed 1; at seed 1 the other rises (+2.3 and +3.0) are about the size of two samplings of the untrained model on the same partition (43.2% at $k = 4$ against 40.7% at $k = 8$), and the replicate and the 1,024-limit run do not rise at all. The cap-share part is 12–30% of the decline in the three original seed-1 runs (−2% and 5% in the replicate and the 1,024-limit run) and 52–72% at seed 2; the cap-accuracy part is small (−1.1 to +1.5 points). The finished-accuracy shares have wide intervals, and they split by seed (seed-1 runs 62–89%, seed-2 runs 34–59%); since seed sets the partition, this may be partition variation rather than run-specific anatomy. What the table supports is qualitative: at 5×10^{-4} both parts contribute, in a balance that differs by partition. At 10^{-4} and 2×10^{-4} the cap-hit share does not rise, the finished-accuracy part exceeds the whole (smaller) change in four of the five runs and is 93% of it in the fifth, and matched finished accuracy falls by 0.6 to 4.7 points with intervals including zero. Termination therefore contributes materially to the decline only at the highest rate, and clearly only at one seed (at 10^{-5} the declines are too small to divide reliably). The long generations that remain open as ordinary solutions and then drift into loosely related prose (e.g. "... acknowledged hurdles acknowledged setbacks ..."), with low exact repetition (mean word 5-gram repeat ratio about 0.02, similar to other generations, though the decoding repetition penalty of 1.1 suppresses exact repeats by itself); this description rests on inspection of examples, not on a systematic annotation. Train-on-gold is different: responses shorten by 55–59%, cap-hit falls from 43% to 7–8%, and finished accuracy collapses (−25.6 and −25.9 points matched). This collapse may owe something to the training recipe rather than to the data alone: the loss is taken over the prompt as well as the response, and reference solutions are short and terse, so even one epoch on 350 of them per round can pull the model far from its own reasoning style; we did not test a response-only loss.

Two checks of the pipeline. The training limit (`max_length`= 512 tokens, prompt included) equals the generation cap, so 24–26% of the kept pool (108 of 440 examples in Run A at round 0) is truncated in training: it loses its final answer and end-of-sequence token and could teach the model not to stop. We reran Run A at seed 1 and 5×10^{-4} once with a training limit of 1,024 tokens, together with a matched replicate that keeps the 512-token limit. Removing the truncation changed little in this one run: the round-2 cap-hit share is 43.8% with the longer limit, 43.0% in the replicate and 46.2% in the original Run A, and matched finished accuracy falls by 5.7, 5.0 and 9.1 points. The differences between the three arms are within the spread of the one same-configuration pair we have (the original and the replicate, which are themselves two of the three arms, 4.1 points apart matched), so this pair is a poor estimate of replicate noise; and the longer limit does not remove every unterminated example: 43 of the 440 accepted

Table 13. Adapters re-scored on the same exam problems (seed-1 partition, $k = 4$, 150 problems) under a 512- and a 1,536-token generation cap. Round-2 adapters from runs at 5×10^{-4} ("replicate" keeps the 512-token training limit, "1024 limit" trains with no truncated example) and 2×10^{-4}, against the untrained model. One adapter per row; re-scoring the same adapter moves pass@1 by 2.7–6.7 points against the run's own round-2 value. Only summary statistics were saved for these re-scorings, so no bootstrap intervals and no cap-hit counts are available; "Degenerate" and "Acc. finished" use the character proxy of Section H, which undercounts cap hits, and are shown only for completeness. Sample accuracy rests on 600 samples clustered in 150 problems.

Model	Cap	pass@1	pass@4	Sample acc.	Degenerate	Acc. finished
Untrained	512	36.0%	52.0%	36.3%	8.7%	39.2%
	1536	40.7%	59.3%	43.2%	2.5%	44.1%
5×10^{-4}, replicate	512	29.3%	46.7%	28.3%	30.0%	38.1%
	1536	35.3%	52.0%	34.2%	23.0%	43.3%
5×10^{-4}, 1024 limit	512	26.0%	45.3%	27.5%	37.2%	40.8%
	1536	30.7%	50.7%	29.5%	32.7%	41.3%
2×10^{-4}	512	27.3%	48.7%	28.5%	28.7%	37.9%
	1536	32.0%	52.7%	32.0%	23.7%	40.8%

round-0 examples (9.8%) are cap-hit generations, which have no end-of-sequence token under any training limit. With one run per arm this can only exclude a truncation effect much larger than the 4-point gap between the pair. The replicate reproduces the direction of Run A's decline (pass@1 $33.3 \to 31.3 \to 25.3\%$, pass@4 $50.0 \to 48.0 \to 44.0\%$).

Re-scoring the same adapters under a 1,536-token cap, with everything else fixed (Table 13), is consistent with the 512-token cap not being the whole story, though with one adapter per row and no intervals it does not establish it. The untrained model gains *more* from the larger cap than the adapters do (sample accuracy +6.8 points, against +5.8, +2.0 and +3.5), so the adapters' deficit does not close: −8.0, −8.8 and −7.8 points at 512 tokens and −9.0, −13.7 and −11.2 at 1,536. This does not separate less accurate solutions from generations that do not stop: at 1,536 tokens the proxy's "Acc. finished" is close for all four rows (40.8–44.1%) while its "Degenerate" share is 2.5% for the untrained model and 23–33% for the adapters, which points to non-termination, though the proxy undercounts cap hits. Since re-scoring the same adapter moves pass@1 by up to 6.7 points, differences of one to three points between rows, including the comparison of gains, are not interpretable. The data suggest that the gap persists at a tripled cap, whether from less accurate solutions or from generations that do not stop, for single adapters on one partition; they do not show what a still larger cap would do.

I Discussion in Full

I.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. On held-out problems no run raises pass@1 or pass@k by an amount whose interval excludes zero, at any of four learning rates, and the sample-accuracy intervals exclude gains larger than about 2.2 points. This bound is loose on three counts: it is read off the observed intervals, not a pre-specified equivalence test; it uses sample accuracy, which we chose as primary after the runs (on the pre-set pass@1 and pass@k the upper ends reach +8.0 and +7.3); and the intervals omit training-seed variance, which is of the same size (a replicate of Run A at 5×10^{-4}, seed 1, ends 3.2 points of sample accuracy away). In every MATH-500 run except 10^{-4} seed 3 (16/14) the problems lost over the rounds equal or outnumber the problems gained (Table 4; 10^{-5} runs: 20/21 to 32/34 over four transitions), which restates the sign of the pass@k change rather than adding evidence. Most per-problem movement is churn: about 2 to 25 problems are gained and about as many lost per round even where the averages barely move.

Does it learn to write what the verifier accepts? The data cannot settle this. The pattern fixed in advance for this reading (pass@1 up, pass@k flat, style converging) requires pass@1 to rise, so it cannot fire in runs where accuracy falls, and pass@1 never rises detectably (the largest rise is +0.7 [−6.7, +8.0], at 10^{-4}, seed 1); the rule is one-sided and its silence is not evidence against verifier-style learning. Its style leg, the pre-set `\boxed` share, was evaluated on its own and also fails: in self-training the share does not converge but falls (all 2,000 generations per round: 52% → 36% and 54% → 37% at 5×10^{-4}, Table 11; the 600 exam generations: 55.5% → 38.3% and 58.5% → 42.5%). The post hoc refinements do not settle it either. The kept pool (89–91% boxed at round 0) is less boxed than the *finished* exam generations it is compared with (95.6% and 98.5%), and the finished share falls (69.7% and 82.9%) as the pool's does (73–77% by round 2). That is what imitation of the pool would predict as well as what drift would, so the share does not discriminate. The post hoc false-reject audit (Section 5.3) finds no adaptation to the gold-storage defect once the falling `\boxed` share is accounted for, but it has little power. The `\boxed` share is also not a clean measure of what the verifier prefers, because the gold-storage defect (Section B) penalises some `\boxed` answers. The one strong style convergence (`$` spans 5% → 98%, `\boxed` 52% → 86%) happens under train-on-gold, with no verifier in the loop and with falling accuracy: style converging alone, which by our rule says nothing about verifier-style learning.

What does change? In this post hoc description, most clearly at the highest rate, the model gets less accurate on the solutions it does finish. The matched intervals exclude zero in five of six MATH-500 runs at 5×10^{-4}, in both GSM8K runs (10^{-5} and 5×10^{-4}) and, on MATH-500 at 10^{-5}, only just, for A and B at seed 0, which are close to one replicate; at 10^{-4} and 2×10^{-4} the point estimates are negative but the intervals include zero (both GSM8K figures are single runs). More generations run into the token cap only at 5×10^{-4}: by 2–14 points in the

four verifier-filtered runs and two no-filter controls on MATH-500, with intervals excluding zero at seed 2 and for one run at seed 1, and not in a replicate; the change in cap-hit share accounts for 12–30% of the decline in the original seed-1 runs and 52–72% at seed 2 (Table 7). At lower rates the cap-hit share changes by -3.3 to $+1.1$ points. The untrained model already hits the cap in 41–48% of MATH-500 generations, so the cap limits every round.

What happened to the pooled collapse? It does not reproduce when each dataset gets its own adapter and is scored on held-out problems. Because the pooled design also trained on about 4.5× more examples per round, and our higher-rate runs suggest (on one seed) that stronger training produces larger declines, which are at least compatible with a dose effect (we did not apply the measures of Section H to Design P, and its acceleration under shrinking pools is not explained by dose, Section D), we cannot say whether pooling, selection-set scoring or training dose produced it. We no longer claim that pooling and selection-set scoring "manufactured" the collapse.

I.2 Interpretation

An earlier version of this work explained the Design P curves with a self-reinforcing shrinking-pool mechanism: the biased verifier accepts a smaller, more style-skewed set each round and the skew compounds. The data give it no support, but they do not rule it out either. The patch removes a severe but rare bias and changes the pool by a handful of examples, so it tests how strong the bias is per answer, not whether a frequent bias compounds; the no-filter control declines without any verifier, as much as Run A at seed 1 and more at seed 2 (sample accuracy -10.2 against -4.7), but it also trains on 3–4× more examples, so its decline is confounded with dose; and the training-dose difference between the designs offers a simpler account that we have not tested directly either. Pool size is also partly definitional (it is sample accuracy on the selection items times the number of samples), so its co-movement with accuracy, which holds on pass@1 in Design I as well (Run C: pool -20%, pass@1 -12.7; Run A: -3%, -2.7), says little about mechanism.

Design I differs from Design P in at least four respects at once (isolation, a held-out split, n per run falling from 1,000 pooled problems to 500, and the training dose), so the P/I contrast is descriptive. The clean comparisons are internal to Design I and its higher-rate counterparts, and even there the controls differ in dose. The defensible statement about the verifier is narrow: *removing a format defect present in under* 1% *of generations, or removing the filter entirely, made no consistent difference at the resolution of this setup.* The one frequent systematic false negative we measured, the gold-storage defect, was present in every run and was never varied, so whether systematic false negatives are harmless in general, as Egashira et al. find for RLVR, our data cannot say.

I.3 Practical Suggestions

These suggestions are not findings of this paper: none was tested here. They are general good practice that problems in our own runs (one 1.5B model, one rule-based verifier, LoRA SFT, at most three seeds per condition) illustrate, and our data do not show that following them changes a conclusion. *Hold out evaluation items and keep one adapter per dataset* if you report per-dataset trajectories, so the curves can at least be read; our data do not rank these factors. *Match training dose across arms*, or report optimizer steps: pool size changes with filtering, k and dataset, and with one epoch it sets the number of updates. *Check that the training step changes the model* (loss and output statistics after one round) before reading a flat curve as stability. *Count the prevalence of a suspected verifier defect in the loop's own outputs before building a control around it;* a treatment that changes 0.5% of the pool cannot be detected with 150 problems. *Save token counts and report cap-hit directly,* not a character-length proxy, and adjust any "accuracy on finished solutions" for which problems still finish; our proxy misread a change in text style as a change in termination. *Report intervals,* and track pass@k and per-problem gains and losses alongside pass@1.

J Limitations in Full

1. *Unequal training doses.* With one epoch and batch size 2, optimizer steps are half the pool size: round 0 takes about 1,981 steps (inferred) in Design P, 436 in Run A and 951 in Run C at 10^{-5}, 220 in Run A at 5×10^{-4}, 700 for no-filter and 175 for train-on-gold. Every cross-arm comparison mixes the intended variable with dose, and the cause of the Design P decline is not identified.
2. *Underpowered controls.* The patch changes 0–4 training examples; at 10^{-5} the adapter also changes the outputs little. Its null is expected whatever the truth, and we report no equivalence bound. The no-filter and train-on-gold controls have two seeds each.
3. *Intervals cover only part of the uncertainty.* The bootstrap intervals cover problem draw and generation sampling, not training-seed variance. Most conditions have two or three seeds; the following have one: Run C at both rates, the truncation check (one replicate pair at 5×10^{-4} and one run with a 1024-token training limit), and each token-cap re-scoring. pass@1 is a single draw per problem.
4. *Learning rate.* The original 10^{-5} was an untuned default; the evidence that it changes outputs little is behavioural (unchanged length and cap-hit share), since its loss does fall. The higher-rate grid is coarse and uses three rounds and $k = 4$.
5. *Cap-hit is reconstructed:* token counts were not saved, so we re-tokenise the decoded text and count a generation as a cap hit at ≥ 508 of 512 tokens. The token-cap re-scorings saved only summary statistics, so their termination figures still rest on the character proxy, which we show undercounts cap hits.

6. *Verifier measurements.* The false-negative audit used other models' solutions and a scripted correctness check, inspected only rejections (so no false-positive rate). The gold-storage defect was measured afterwards on the loop's own outputs by re-scoring (Section B); it lowers absolute pass@1 by 4.7–8.0 points in the three self-training runs we re-scored (other runs are extrapolation), was present in every run including both arms of the patch control, and was never varied in a training run. It overlaps with the unparseable-gold correction (46 of the 51 unparseable golds become parseable once boxed), so the two corrections must not be added. The re-scored false-negative share assumes the re-scored verifier makes no false accepts.
7. *Restart-from-base design and scope.* A fresh adapter on the base model each round leaves compounding little room, so the absence of compounding is partly a property of the design. One model size in Design I, one verifier, one training method, at most five rounds; full fine-tuning, continued training and online RL are untested.
8. *The selecting verifier also grades.* The same math-verify configuration that chooses the training data scores the exam, so any systematic verifier error enters both selection and measurement. The gold-storage re-score (Section B) and the string audit of the train-on-gold runs bound part of this, but only for five runs and one defect class.
9. *A binding token cap.* At 512 tokens the untrained model already fails to finish 41–48% of MATH-500 generations (depending on the seed's exam split), so every measure is taken in a regime where length and accuracy are entangled, and the 512-token training limit truncates about a quarter of the kept pool. Our conclusions are about this regime; a larger cap during the loop is untested.
10. *No equivalence margin and many intervals.* We did not fix in advance how small a gain would count as "no gain", so the 2.2-point bound is read off the observed intervals. Table 5 alone holds 76 intervals and the text reports about ten more, none with a multiplicity correction. They are not independent (they share runs and problems), so "about four would exclude zero by chance" is only a rough guide, but single intervals that only just exclude zero should be read cautiously.
11. *Possible test-set contamination.* MATH-500 and GSM8K are public and may be in Qwen2.5's training data. This would raise absolute accuracy, and it need not affect all arms or rounds alike: if training on self-generated solutions makes the model forget memorised items, part of a within-run decline could be loss of memorisation rather than of skill. We did not test it.
12. *Training recipe.* The loss covers the prompt as well as the response, and we tried one LoRA configuration (rank 16 on query and value projections). Both could matter, most plausibly for the train-on-gold collapse, and neither was varied. No arm in any design improves detectably on its round 0, including training on reference solutions, so we have no positive reference showing that this recipe can help a 1.5B model at all; the declines may partly reflect the recipe rather than self-training.

13. *Limited power.* With 150 exam problems the smallest changes the design would detect with 80% power are about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy (Section 3.4).
14. *A weak adaptation audit.* The false-reject audit of Section 5.3 was defined after the runs, covers one defect class whose effect is bound to 54–62 problems (only 24–28 of them escapable by any answer form), rests on 2–12 training examples per round that could teach the avoidance, and has no positive control showing it would detect adaptation that did occur.
15. *Confounded learning-rate grid and one model family.* The higher rates also use $k = 4$, three rounds and smaller pools, so only comparisons among them isolate the rate. All runs use Qwen2.5 models, whose behaviour under RL with spurious rewards is known to differ from other families [9].

K Future Work

Separate the explanations of the pooled decline with a pooled run scored on a held-out split and an isolated run matched in optimizer steps (by fixing the step count per round rather than the number of epochs); save token counts at generation time and test a larger generation cap during the loop, since the untrained model already hits 512 tokens in 41–48% of MATH-500 generations; refine the learning-rate grid and replicate Run C; and run a positive control (a verifier biased against a frequent format) to show what effect size this design can detect. For the false-reject audit, a positive control would inject a defect that rejects a frequent, avoidable answer form (for example all `\boxed` answers on a fixed tenth of the problems) and check that r_t on those problems falls; without it a flat r_t cannot be read as absence of adaptation.

The gold-storage defect (Section B) is the natural next verifier control. We implemented and unit-tested a patch that wraps the gold in `\boxed{}` whenever the prediction contains one and the gold does not; re-scoring shows that in a training run it would enlarge the round-0 kept pool by about a fifth ($440 \to 542$ and $872 \to 1{,}066$), roughly 34–49 times the change made by the `\dfrac` patch, so unlike Run B such a run would actually perturb the training data. Because a larger pool also means more optimizer steps, the comparison should hold dose fixed, either by subsampling the patched pool to the unpatched size or by fixing the step count. Given that same-configuration runs end several points apart, it needs several seeds per arm; a single seed would detect only a large effect.